Critical Design Review Presentation

Team VALOR

AEROSPACE ENGINEERING

Team Organization

Abby Rajagopal
Team Lead

Camden Hill
Propulsion

Chris Battung
Communications

Ethan Auringer
Crew Systems

Jacob Arnold
Structures

Lucas Schuh
GNC

Roberto Valdovinos
Power

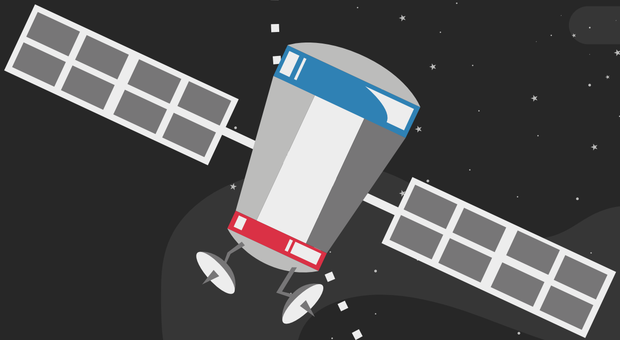
Hunter Rogers
Avionics

AEROSPACE ENGINEERING

NORTHROP GRUMMAN

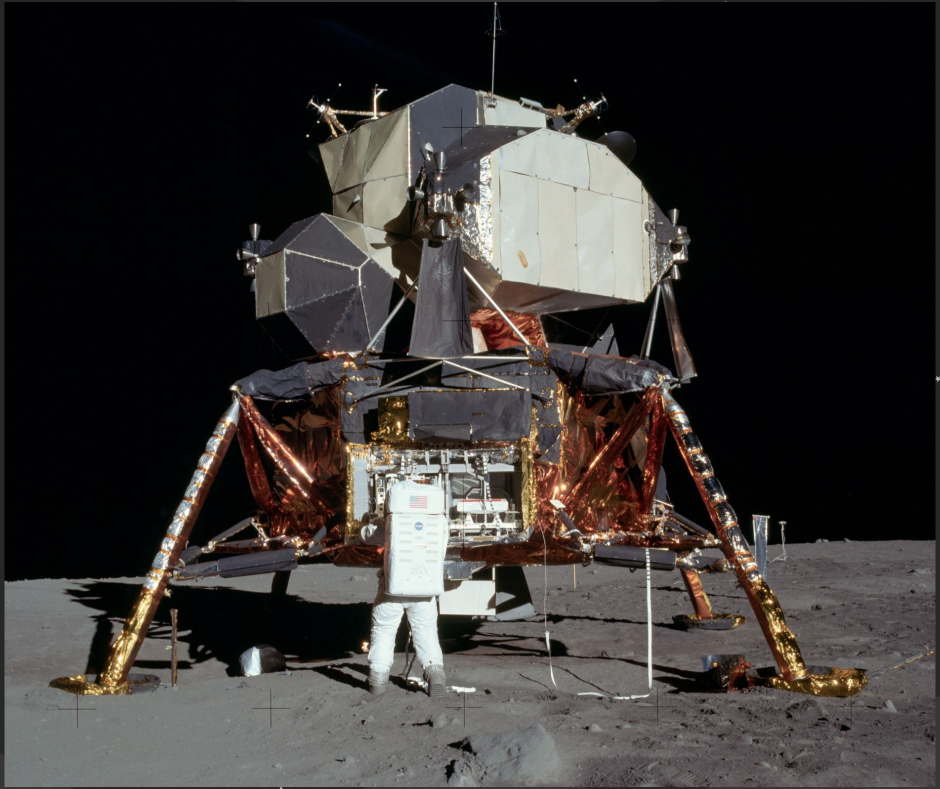
01

Mission Overview



Background & Motivation

- The Apollo LM was the first AND only crewed space vehicle to land on the Moon
- Originally designed for short equatorial mission up to 3 days
- Legacy systems are now out-of-date due to technological advancements
- Goal is to expand capabilities and maximize performance for the Artemis age and beyond



Stakeholder Needs

Stakeholders: Northrop Grumman, Professor John Connolly, HLS Program (NASA), FAA

Phase 1 Objectives

- Analyze the Apollo Lunar Module systems and subsystems
- Determine where 21st-century technology upgrades can be integrated
- Track Master Equipment List (MEL) and Power Equipment List (PEL)
- Communicate changes to design, mass and vehicle capabilities

Phase 2 Objectives

- Determine redesigns that can be implemented to expand mission scope/profile
- Identify and build or simulate a system/hardware that can be tested
- Provide lessons learned to drive future vehicle development

Phase 1 Summary

Communications: Increase transmit & receive data capabilities
GNC: 86% decrease in subsystem mass with increased performance
Structures: Use of new lightweight materials allow a subsystem mass decrease of 17%
Propulsion: Selected Selected Methalox rocket engines to greatly reduce propellant mass
Crew Systems: Established Cabin Atmosphere of 34% Oxygen at 8.2 psia and Updated the CO2 Scrubbing and Cooling
Power: Reduced mass by 60% and incorporated rechargeability
Avionics: Added Redundant Systems and Increase Computing Power

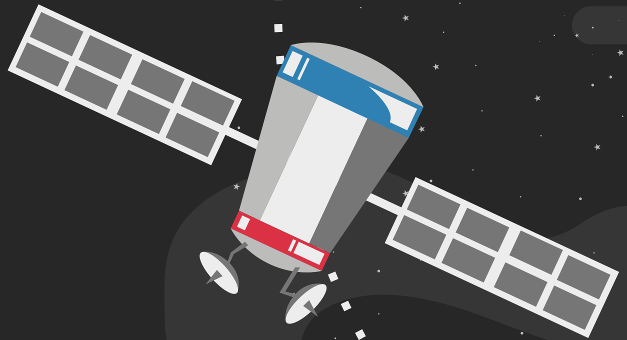
Apollo LM
Total Mass: 15,103 kg
Total Power: 73 kWh

→

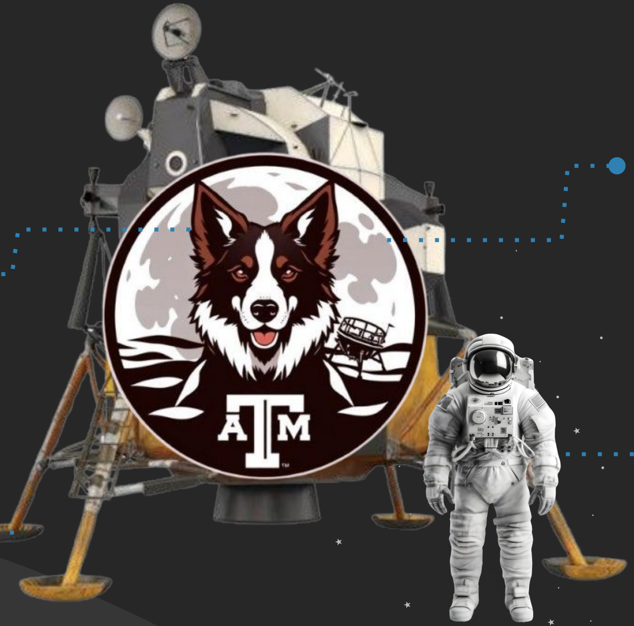
Phase 1 LM
Total Mass: 9752 kg
(43.1% mass reduction)
Total Power: 155 kWh

02

System Concept & Architecture



Redesign Needs



Capable of longer and multiple trips

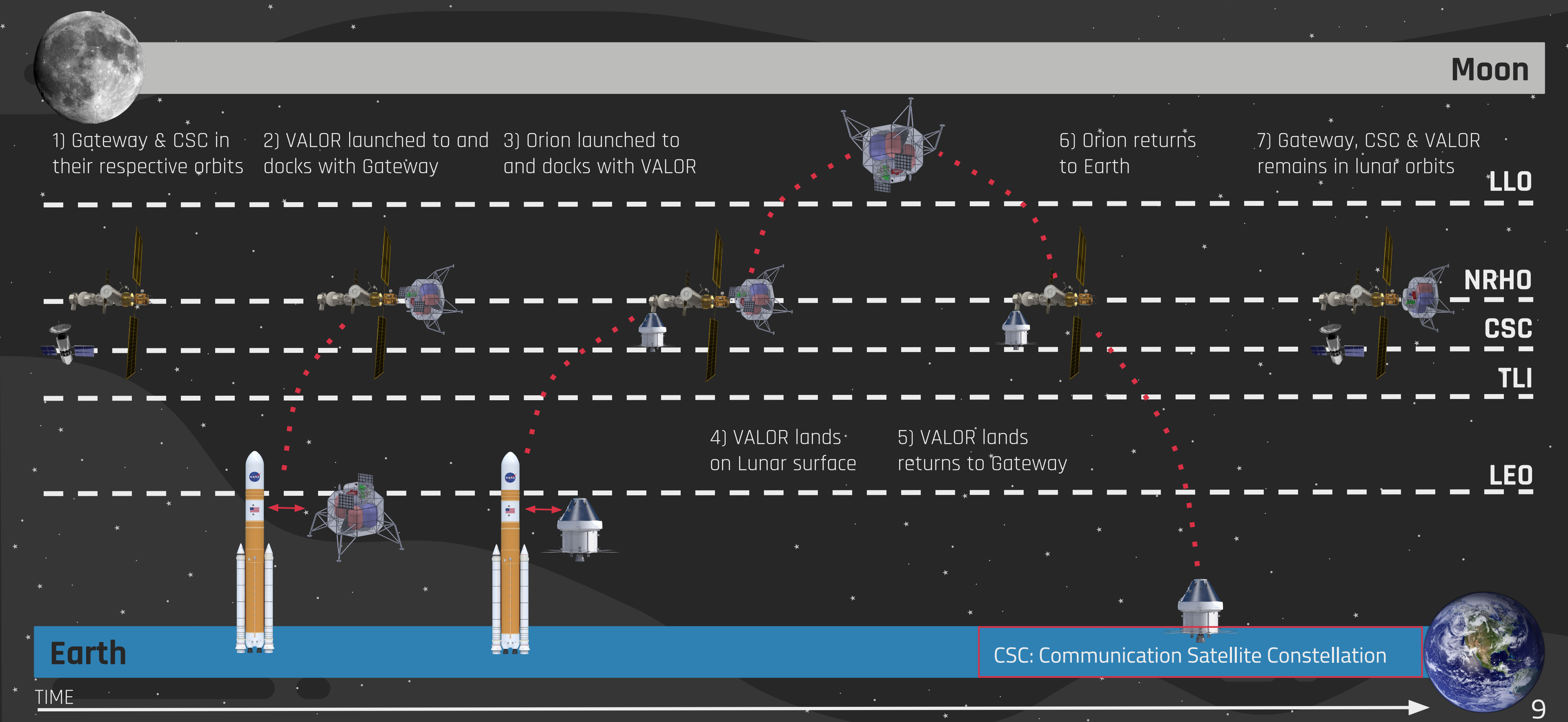
Have continuous connection with ground

Incorporate large crew cabin

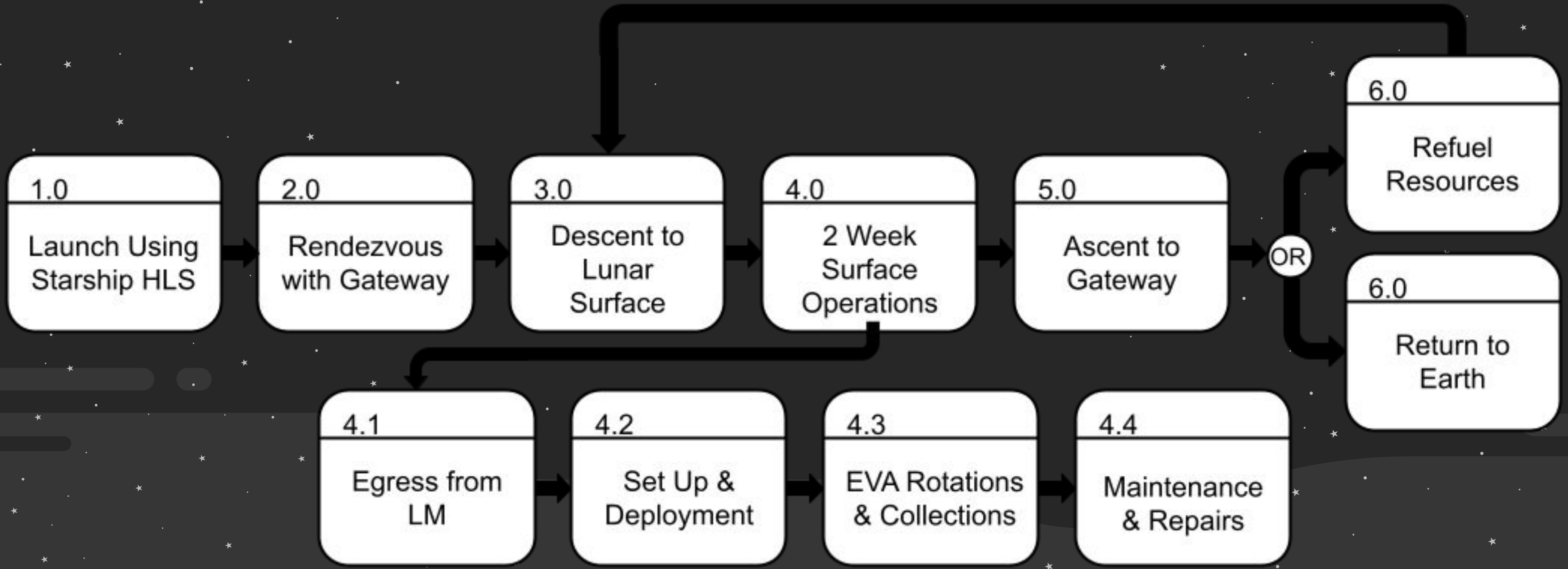
Travel to diverse lunar regions

Presenter: Abby Rajagopal

Concept of Operations

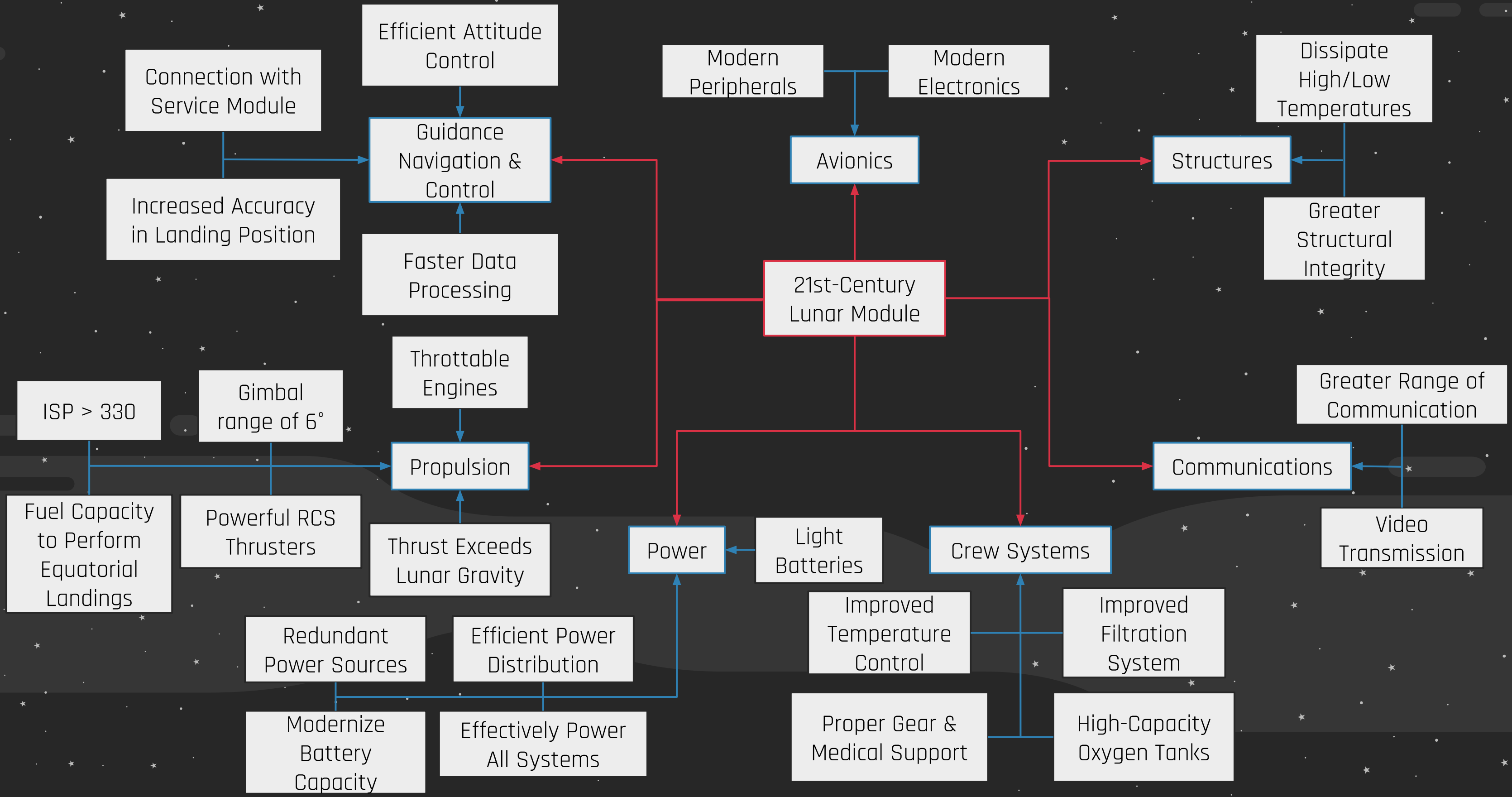


Functional Flow Diagram



Presenter: Abby Rajagopal

System Decomposition



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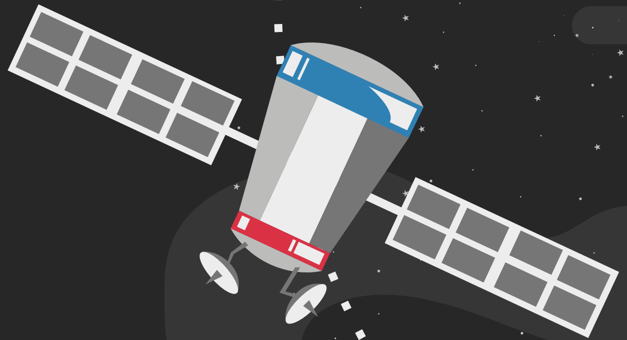
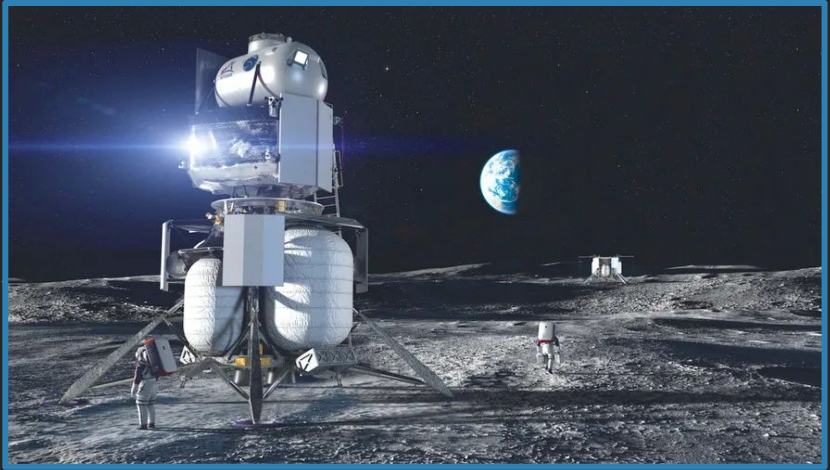
System Design



Why HLS?

- **Design Relevance:** following HLS requirements allows us to be consistent with Artemis, a current NASA architecture
- **Industry Alignment:** multiple commercial partners under contract to compare and reference designs
- **Mission Driven:** Ideal for developing concept of operations and test scenarios



System Goals

#	Goal	Rationale
G.1	Deliver four person crew to the Moon	Ensure crew safely reaches lunar surface
G.2	Support two week missions	Evolve into a safe and affordable long-term approach to a sustained presence
G.3	Support expanded crew and cargo capacities	Support recurring operations by allocating for more astronauts and payload

System Requirements

#	Requirement	Rationale	Traceability
R.1	VALOR shall maintain charge/re-charge states adequately to accomplish stated mission objectives	Status of power related measurements should demonstrate that parameters values are within acceptable levels and the sensors are measuring in a healthy manner.	G.2, G.3
R.2	VALOR ECLSS system parameters shall be within model predictions given the measured VALOR environmental state	Status of ECLSS related measurements shall demonstrate that parameters values are within acceptable levels and the sensors are measuring in a healthy manner	G.2, G.3
R.3	Communication data shall demonstrate that VALOR has established communication with the ground and Gateway (if applicable) and HLS is transferring data appropriately considering various orientations	Status of communication-related parameters shall demonstrate that values are within acceptable levels	G.2
R.4	VALOR avionics system parameters shall be within operating parameters and the sensors are measuring in a healthy manner	Status of avionics sensor parameters shall demonstrate that values are within acceptable levels and measuring in a healthy manner	G.2
R.5	Thermal systems shall demonstrate that VALOR system parameters are within model predictions given the measured HLS environmental state	Status of thermal system related measurements shall demonstrate that parameters values are within acceptable levels and the sensors are measuring in a healthy manner.	G.2, G.3
R.6	VALOR shall demonstrate the ability to reorient and translate in support of rendezvous, landing and return operations	Status of propulsion system related measurements shall demonstrate that parameters values are within acceptable levels and the sensors are measuring in a healthy manner	G.1, G.2
R.7	VALOR shall demonstrate the accuracy of the Guidance and Navigation Control data required to achieve MPS and RCS control modes	Status of navigation and control related measurements shall demonstrate that parameters values are within acceptable levels and the sensors are measuring in a healthy manner	G.1, G.2
R.8	VALOR shall demonstrate sufficient resources to accommodate a minimum duration mission and return to Gateway or Orion	VALOR should safely support crewed flight operations shall be demonstrated by assessing vehicle telemetry for habitable conditions and relevant HLS data for safe operations	G.2

Top Level Requirements

I.D.	Subsystem	Requirement
R.1	Communications	The LM shall transmit visual, audio, and lifeline data to the CSM at a distance of 30,000 km.
R.2	GN&C	VALOR shall be able to land on the lunar surface from multiple orbits
R.3	Structures	VALOR structure shall be reusable
R.4	Propulsion	VALOR shall have sufficient propellant to repeat Apollo 11's Mission
R.5	Crew Systems	VALOR shall serve as a living space for 4 crew members throughout the duration of the mission
R.6	Power	VALOR shall ensure full subsystem power functionality throughout mission duration
R.7	Avionics	Valor shall include sensors to ensure optimal living conditions

Communications

Communications

Traceability	Requirement	V&V
R.3	VALOR shall transmit visual, audio, and lifeline data to the CSM at a distance of 30,000 km.	Analysis
R.3	VALOR's communication system shall transmit data at a rate of 50 MBPS.	Analysis

Communications

Advantech Wireless AWMT-3000K Transceiver

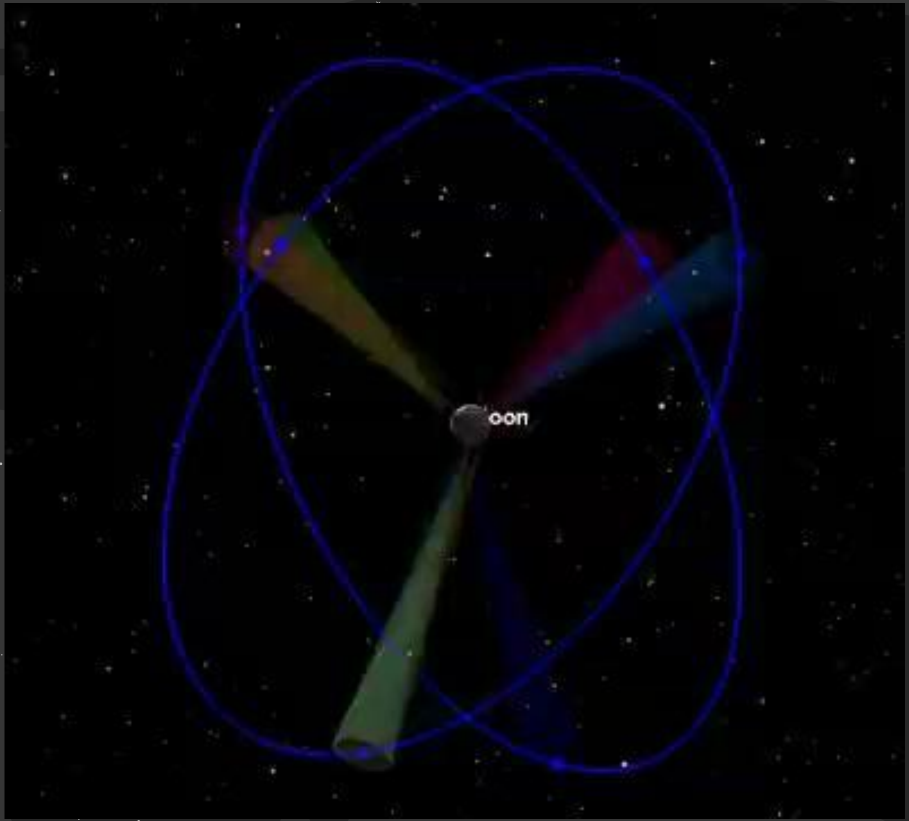
L3 Harris HAWKEYE 4 Lite Antenna

KAPPA Optronics SpaceEye SE 320

BAE Systems Ku-Band SATCOM Antenna

Presenter: Chris Battung

Communications

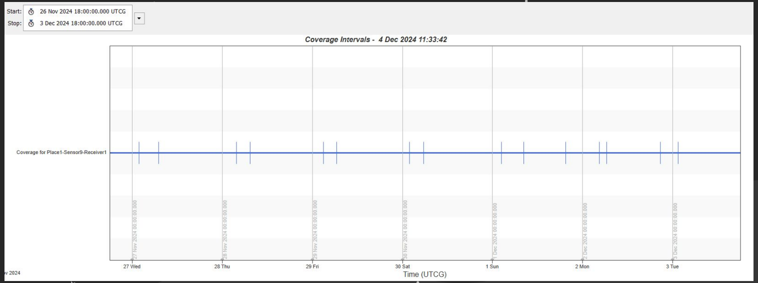


Legend

- Blue lines are the circular orbits of 30,000 km around the moon.
- Blue dots represent the 6 different communication satellites.
- Moving multi colored cones represent the connection between the lunar surface and the satellites.

Results

- All ground points maintain communication with at least 1 of the satellite at all times.



Guidance Navigation & Control

GN&C

Traceability	Requirement	V&V
R.7	VALOR shall safely travel towards the Moon and perform a lunar landing	Analysis
R.7	VALOR shall be able to land on the lunar surface from multiple orbits	Analysis
R.6	VALOR shall know the location of the Gateway at all times	Analysis
R.7	VALOR shall generate possible landing zones based on real-time data	Analysis

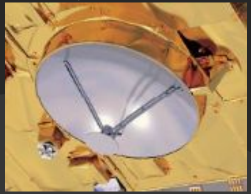
GN&C

Features

- Fully revamped system from hardware to software
- Landing liDAR allows attitude determination from up to 1200 km
- Landing and rendezvous maneuvers can be fully autonomous or pilot-controlled
- Multiple redundant systems to decrease probability of failure

Inertial Measurement Unit (STIM-320)

https://safran-navigation-timing.com/wp-content/uploads/2022/12/STIM320-Datasheet.pdf

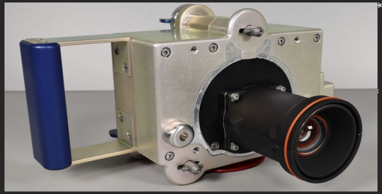


Landing Radar (Poseidon-4)

https://www.eumetsat.int/sentinel-6

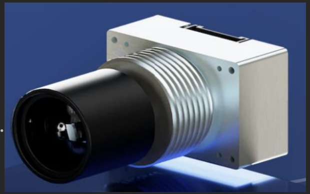
Docking LiDAR (NGAVGS)

https://ntrs.nasa.gov/api/citations/20090033117/downloads/20090033117.pdf



Landing & Docking Camera (DCAM)

https://www.msss.com/files/DCAM.pdf



Star Camera (Cubestar)

https://www.cubespace.co.za/products/cubestar/

GN&C

Landing Region Redundancy

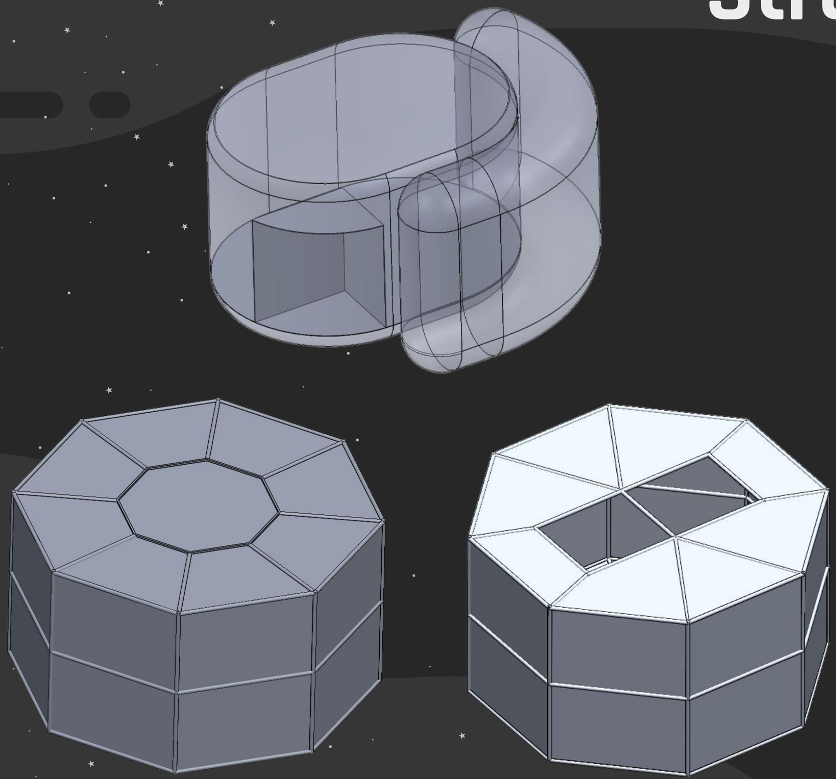
Orbit Propagation Capabilities

Structures

Structures

Traceability	Requirement	V&V
R.6	VALOR shall land on angled planes of up to 15°	Demonstration
R.8	VALOR structure shall be reusable	Demonstration
R.8	VALOR shall have a minimum factor of safety of 2 for critical components	Analysis

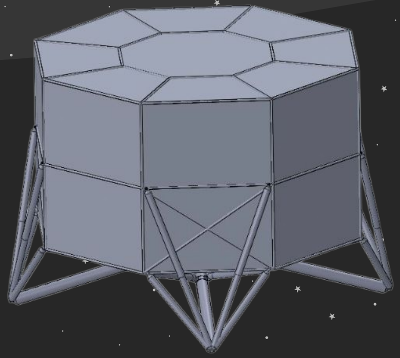
Structures

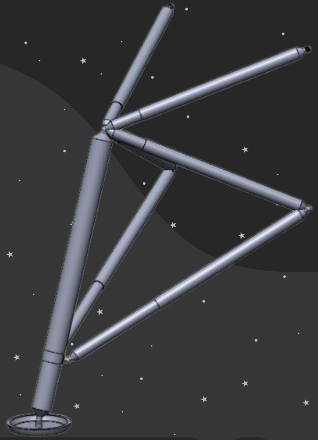


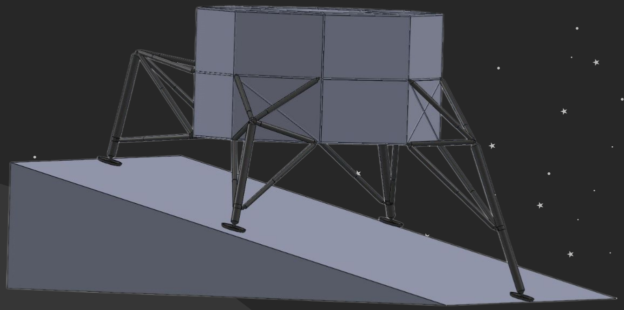
- Large octagonal base for cargo and housing engines and fuel tanks.
- Thin aluminum walls, 3 mm thick, supported by aluminum frame
- Crew cabin and suitport have a curved pressure vessel structure
- Crew cabin has thicker walls, 10mm with HDPE and IMLI insulation layers

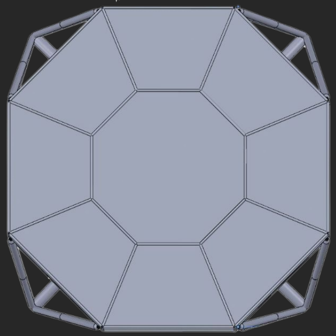
Structures

- CFRP struts
- Foldable landing legs, powered by electric actuators
- Allows for landing gear to fold up, fits into a 8.35 X 8.35 meter footprint
- Supports landing on terrain with up to 15° incline

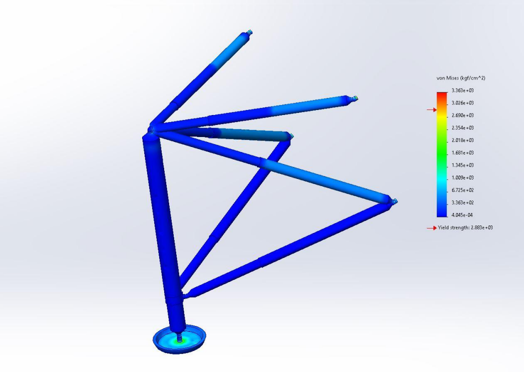


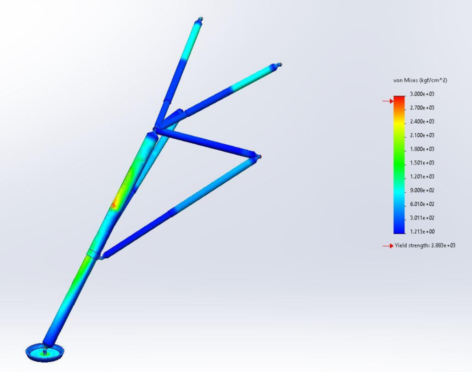


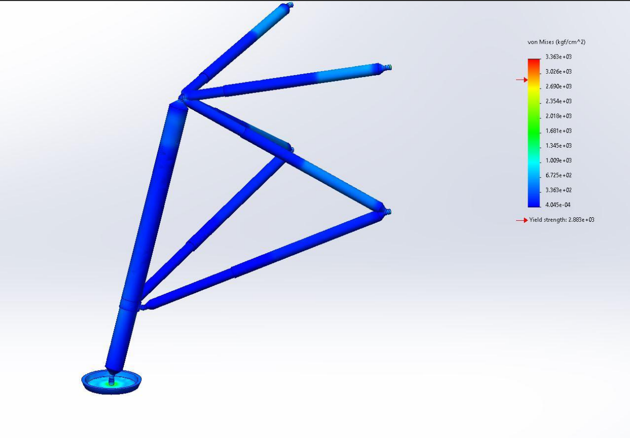




Structures



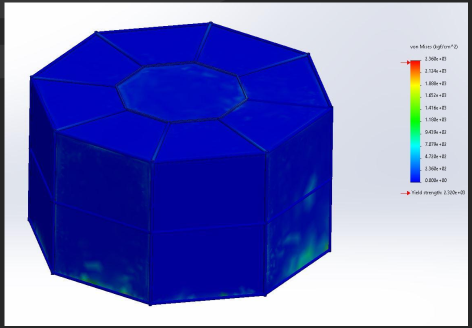


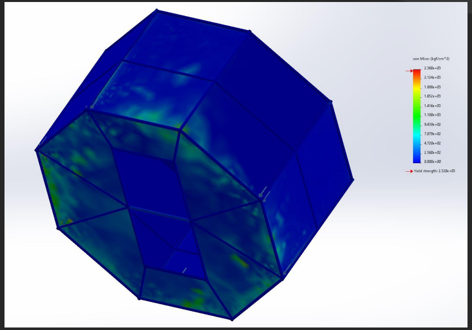


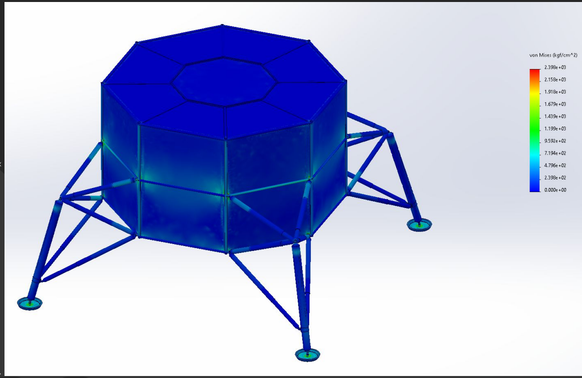
Fixed at wall connection. Loaded with 300kN acting upwards on the footpad.

Fully extended leg (for 15° incline) begins to yield much sooner.

Structures



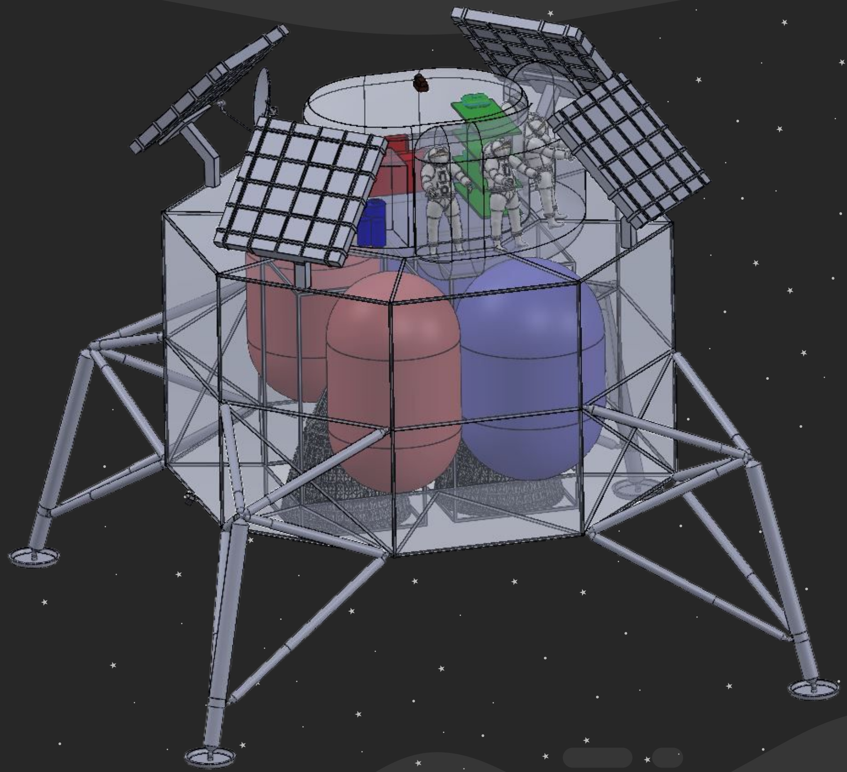




Loaded with weight of crew module on the top surface and weight of tanks and engine on bottom surface in lunar gravity

Structures

- 8.35 meters across
- 10 (8 without landing gear) meters in height
- Total volume approximately 310 m^3
- 70 m^3 available for additional storage in base.
- 40 m^2 area available for additional storage on top.



Propulsion

Propulsion

Traceability	Requirement	V&V
R.6	VALOR shall produce greater than 330 seconds of specific impulse.	Demonstration
R.6	VALOR shall achieve adequate delta-V for a single stage lander that starts at, and returns to NRHO	Demonstration
R.6	VALOR shall produce enough thrust to overcome lunar gravity.	Analysis

Presenter: Camden Hill

Propulsion

- Specific impulse, density & storage temperature
- Choosing Methalox
- Rocket engines for Phase 2
- Methalox RCS thruster design

Phase 2 Trade Study: Commercially Available Methalox Rocket Engines

Engines	Thrust	Mass	Specific Impulse	Throttle	Gimbaling Range	Modifications	Development Time	Development Cost
Custom Design	Moderate	Moderate	Moderate/ High	Moderate	Moderate	Low/ Moderate	Significant	Significant
SpaceX Raptor 3	High	High	High	Effective	High	Low	Low	Low
Blue Origin BE-4	High	Extremely High	High	Effective	Low/ Moderate	Low	Low	Low
NASA Project Morpheus	Extremely Low	Low	Low	Low	Unknown	High	Moderate	Moderate
Intuitive Machines VR900	Extremely Low	Low	Moderate/ High	Effective	Unknown	Moderate	Low	Low
Multiple IM-VR900	Moderate	Moderate	Moderate/ High	Effective	Moderate	Moderate	Low	Low

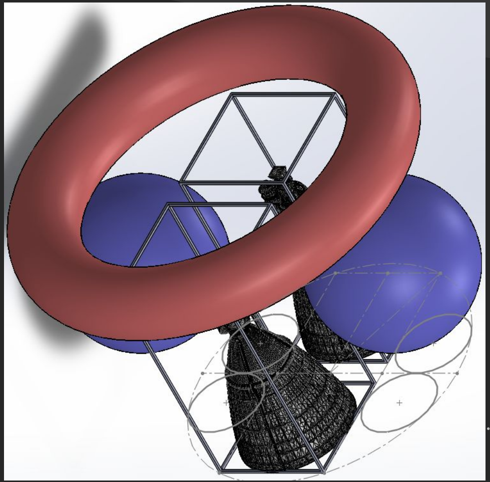
Worse				Better

Presenter: Camden Hill

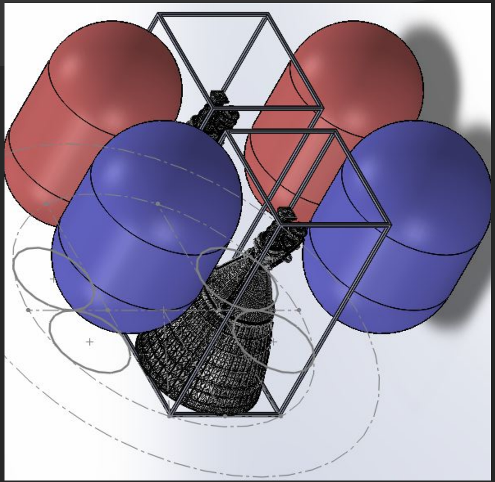
Propulsion

- Consequences of eliminating boil-off using cryogenic cooling
- NASA cryogenic trade study
- A&M COPV carbon fiber thickness
- Toroidal, cylindrical, & spherical tank mass simulation
- Delta-V: ~5000 km/s

Conceptual to Realistic Volume



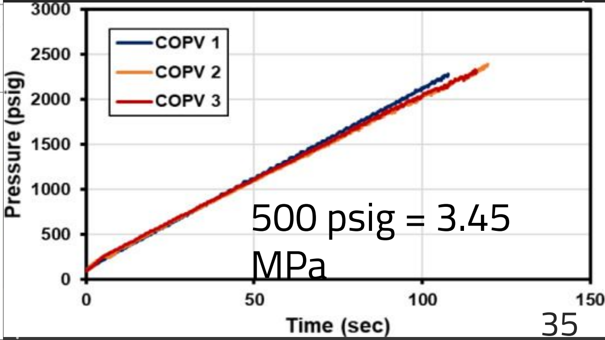
Moving to Cylindrical Tanks



Trade Study: Raptor Engine & COPV Layout and Configuration					
Layout & Configuration	Tank Mass (kg)	Volume Efficiency	Cryogenic Layout	Radiation Protection Mass	Propellant Mass (kg)
2 Oxygen Spheres & 1 Methane Toroidal	980	Moderate	Moderate	Moderate	59,283
2 Oxygen Spheres & 2 Methane Cylinders	864	Moderate	Good	Good	58,957
2 Oxygen Cylinders & 2 Methane Cylinders (Baseline)	808	Good	Good	Very Good	58,795
2 Oxygen Spheres & 2 Methane Spheres	2nd Lowest	Poor	Good	Poor	2nd Lowest
1 Oxygen Sphere & 1 Methane Sphere	Lowest	Extremely Poor	Good	Extremely Poor	Lowest

Worse — Better

Manufacturer	Mass, kg	TRL[a]	Input power, W	Heat leak to fluid, W	Fluid	Phase, Φ	Motor temp., K	Flow, g/s
CryoZone Ciezo	N/A	5	N/A	4	He	Gas (cold)	85	0.6
Sierra Lobo He blower	1	4	0.28	0.14	He	Gas (cold)	85	0.57
Creare NICMOS circulator	1	9	0.89	0.6	Ne	Gas (cold)	80	0.75
CryoZone Noordenwind	N/A	5	0.6	N/A	He	Gas (hot)	300	0.57



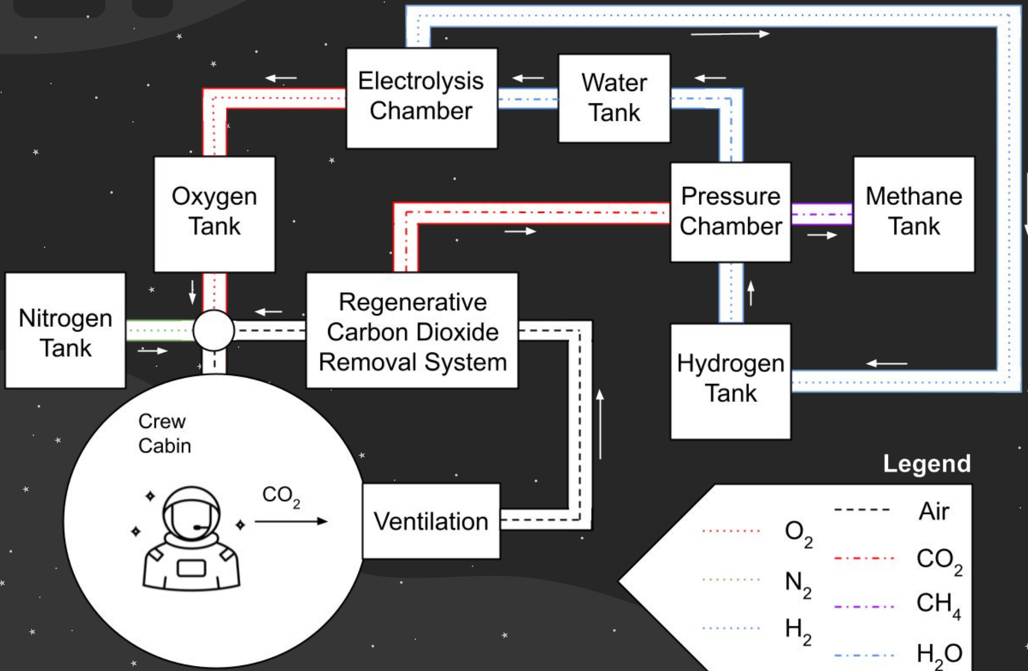
Crew Systems

Crew Systems

Traceability	Requirement	V&V
R.8	VALOR shall have enough water to accommodate crew and cooling for the increased mission duration.	Analysis
R.8	VALOR shall have enough oxygen to support the repressurizations and PLSS refills required for the increased mission duration.	Analysis
R.8	VALOR shall serve as a living space for 4 crew members throughout the duration of the mission.	Testing

Presenter: Ethan Auringer

Crew Systems



Components	With Recycler		Without Recycler	
	Mass (kg)	Avg Power (W)	Mass (kg)	Avg Power (W)
OGS	100	2233.13	N/A	N/A
Sabatier	50	-700	N/A	N/A
RCRS	185	530	N/A	N/A
Lithium Peroxide	N/A	N/A	80.47	N/A
Stored Water	68	N/A	206.8	N/A
Stored Oxygen	37.52	N/A	76.8	N/A
Stored Hydrogen	8.3	N/A	N/A	N/A
Stored Methane	28.14	N/A	N/A	N/A
Tanks	41.22	N/A	38.64	N/A
Totals	**518.18**	**2063.125**	**402.71**	**0**

ΔMass = 115.5 kg

ΔPower = 2kW

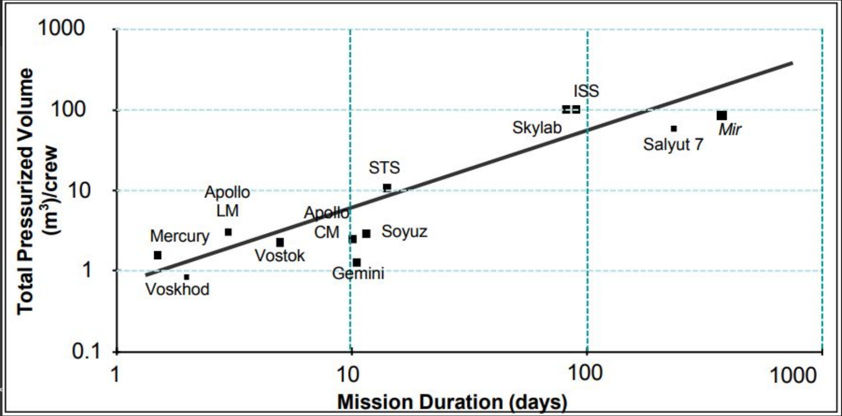
Generation system becomes mass efficient after ~21 days

Crew Systems

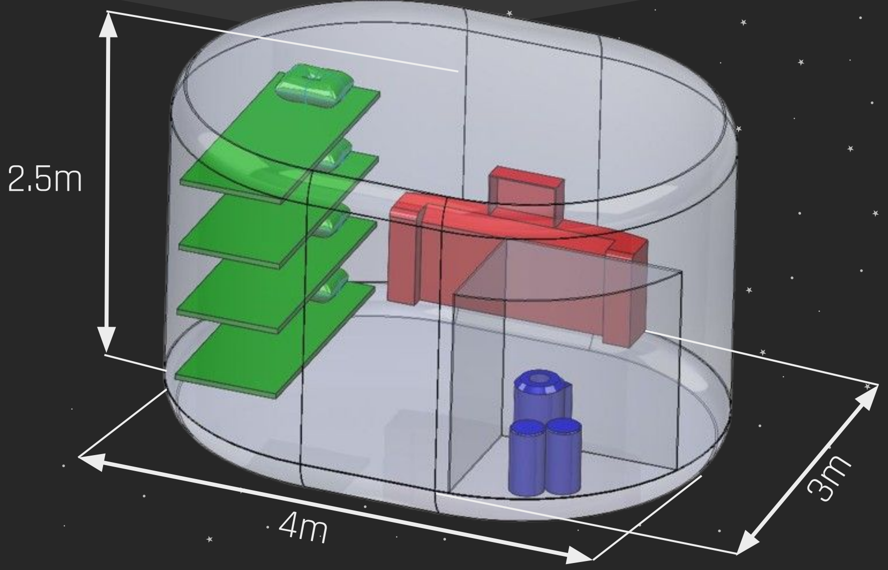
Task-Based Method

Tasks	# of Crew	Vol/Crew	Volume (m^3)
Waste/Cleaning	1	1.7	6.04
Food Prep/Grooming	1	4.34	4.34
Controls/Conference	2	4.34	8.68
Don/Doff	2	6.35	12.7
Total NHV			**31.76**

Experience-Based Method



volume/crew = 6.67xln(duration in days) - 7.79 → 39.25 m^3



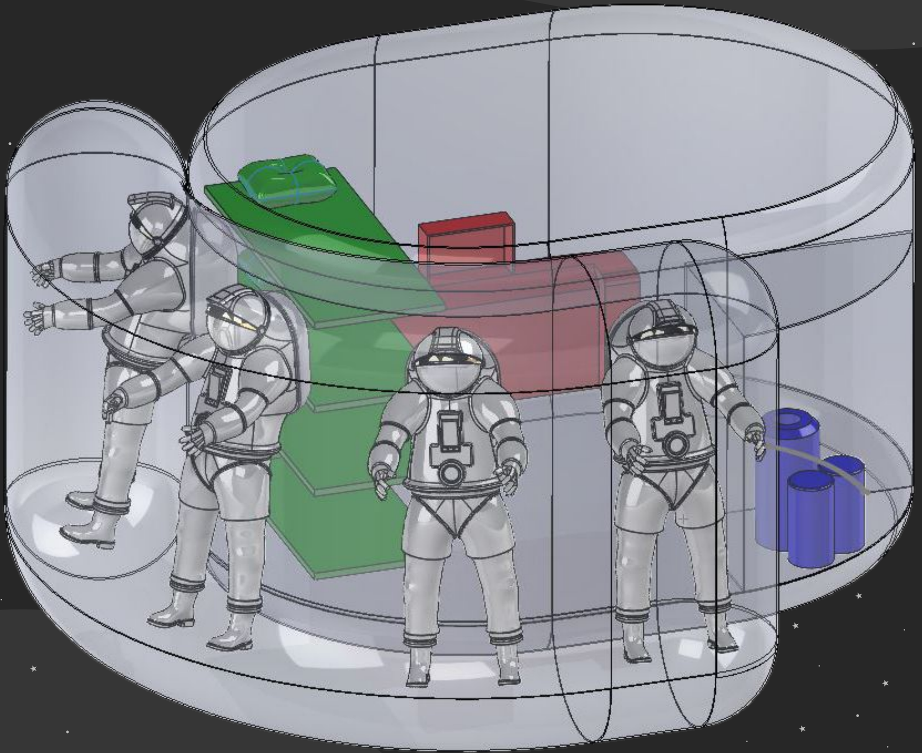
Habitable Volume: 24.4 m^3
Floor Area: 10.1 m^2
Floor Storage Area: 4.5 m^3

*methods are based on NASA-STD-3001

Crew Systems

Main Cabin (24.4m^3)

- Controls
- Conference
- Sleeping
- Food Prep
- Grooming
- Eating
- Waste Management
- Bathing
- Storage



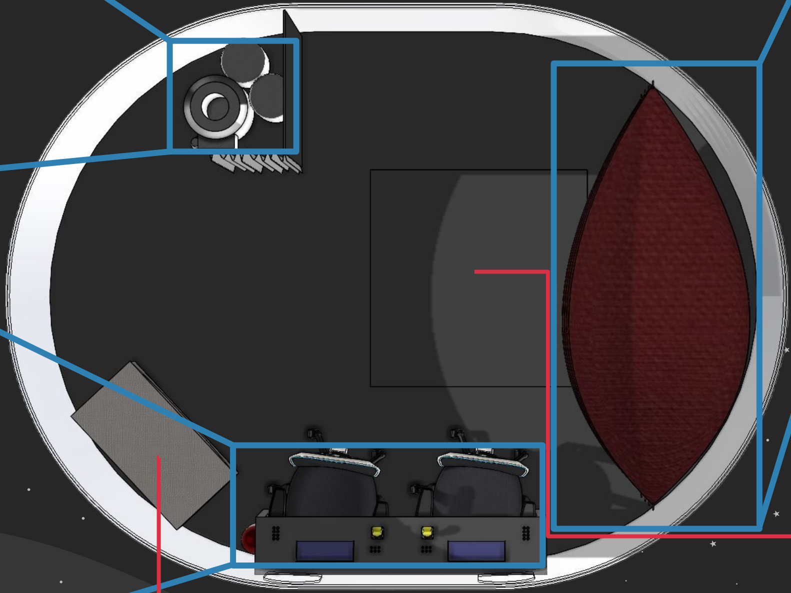
Suitport (13.6m^3)

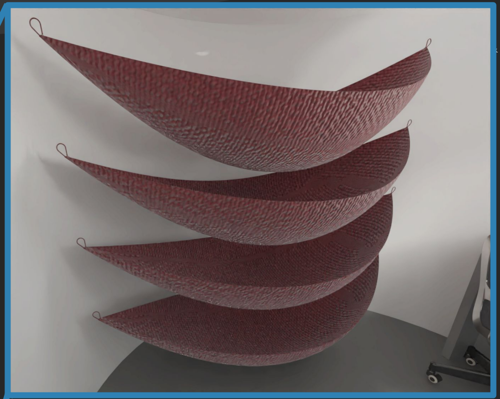
- EMU storage
- Donning and Doffing
- Dust Mitigation

Net Habitable Volume: 38 m^3

Crew Systems

NASA's UWMS

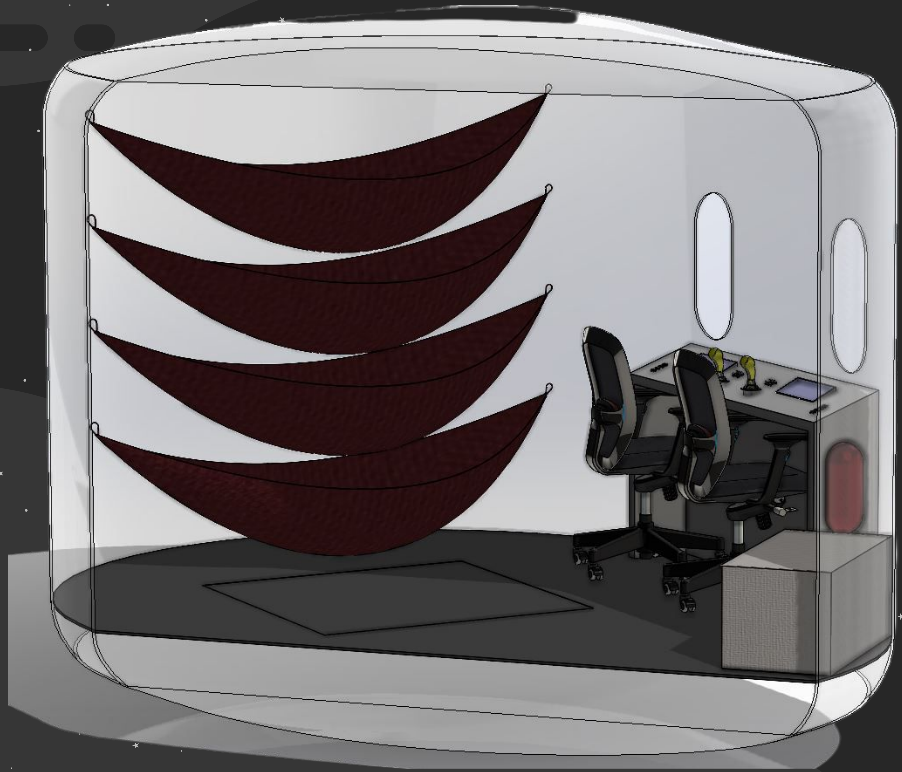


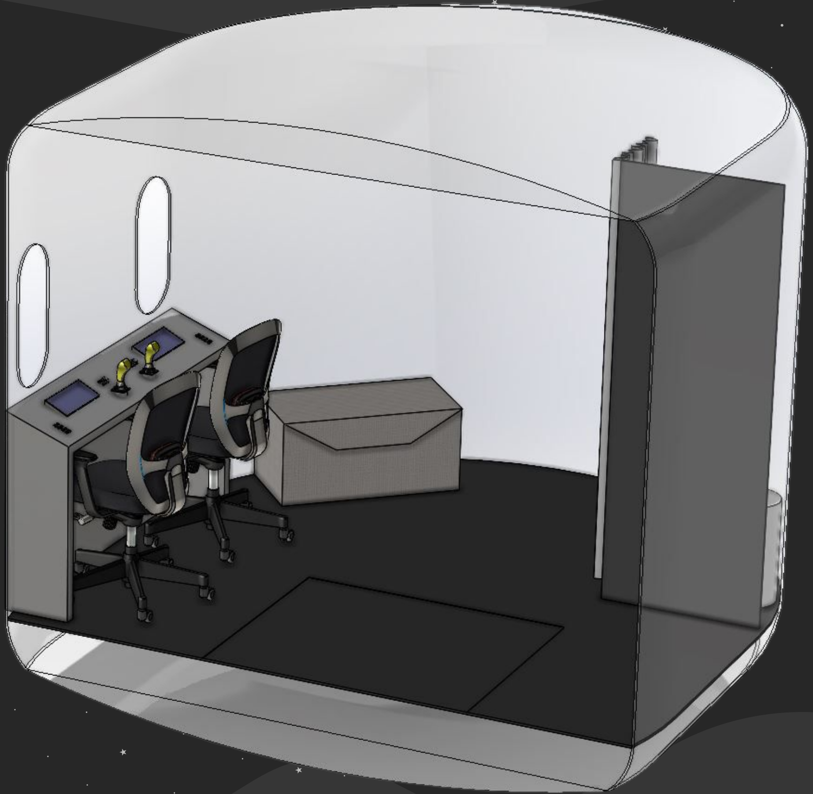


Floor Hatch Space

Food and Personal Storage

Crew Systems





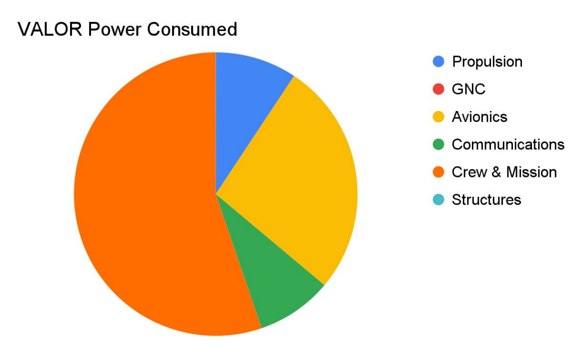
POWER

Traceability	Requirement	V&V
R.1	VALOR shall sustain 50% base power throughout the entire mission	Analysis
R.1	VALOR shall have 3 or more levels of redundancy	Inspection
R.1	VALOR shall have a smart power management system	Demonstration

Power

Stages (Wh)	Propulsion (Wh)	GN&C (Wh)	Avionics (Wh)	Communication (Wh)	Crew and Mission Operations (Wh)	Structures (Wh)
Descent	200	32.8	846.35	492	912.5	10
Surface Operations	200	0.264	574.14	184	1189.5	0
Ascent	200	22.8	846.35	492	912.5	10

VALOR CONOPS Power Consumption



VALOR Power Consumption by Subsystem

Battery	Amprius 500 Wh/kg Battery	Units
Energy Density	500	Wh/Kg
Capacity	50	KWh
Mass	100	Kg
Quantity	3	
Total Weight	300	Kg
Total Power Capacity	150,000	Wh

Amprius Battery Specification

Power

Thin Film Gallium Arsenide Solar Panels	Specifications	Units		Vertical Solar Array Technology (VSAT)	Specifications	Units
Power per Size	400	Wh/m^2				
Mass	2	Kg/m^2		Mass	24.5	kg
Dimension	5	m^2		Dimension	15	m^2
Power Generated per Hour	2380	Wh		Power Generated per Hour	6120	Wh
Quantity	4	#		Quantity	2	#
Mass	40	kg		Mass	49	Kg
Total Power Generated per Hour	9,520	Wh		Total Power Generated per Hour	12,240	Wh
Efficiency	25	%		Efficiency	30	%

Solar Panel and ROSA Specification

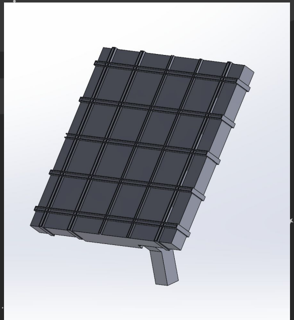
Battery Charge Time Per Power Source	Value	Units
Solar Panel	5.2	Hours
VSATs	4	Hours

Power Source Recharge Time

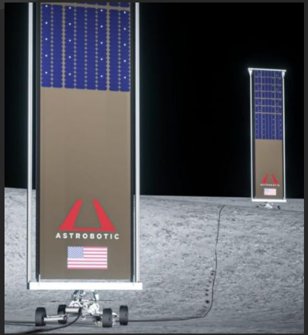
Battery Life per Battery	Value	Units
Decent	80	Hours
Surface Operations	23	Hours
Ascent	80	Hours

Battery Life Span (w/o Power Sources) per Stage

Power



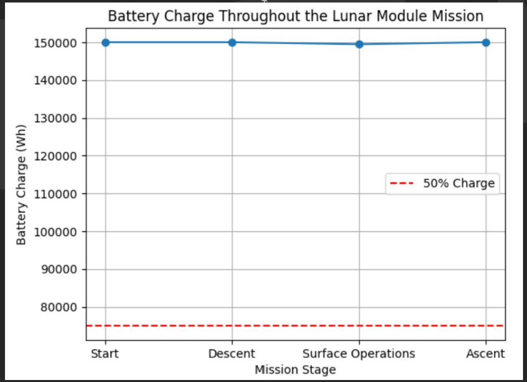
Solar Panel CAD



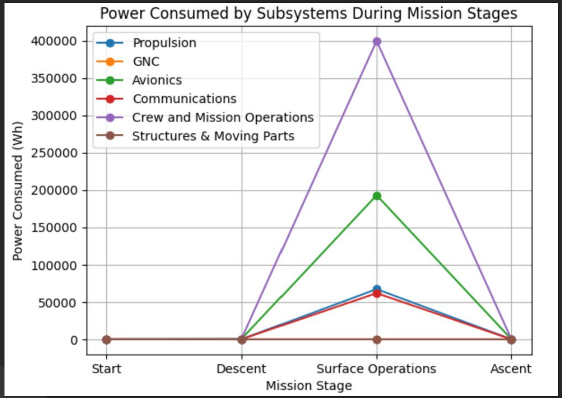
VSATs

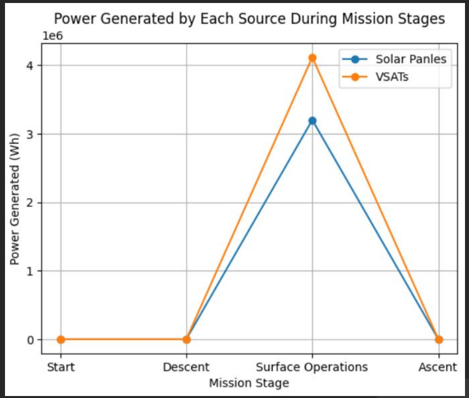
Total Mass (Kg)	Total Power Consumed in 14 days (Wh)
389	722,974

Power Subsystem Mass and Power Consumption



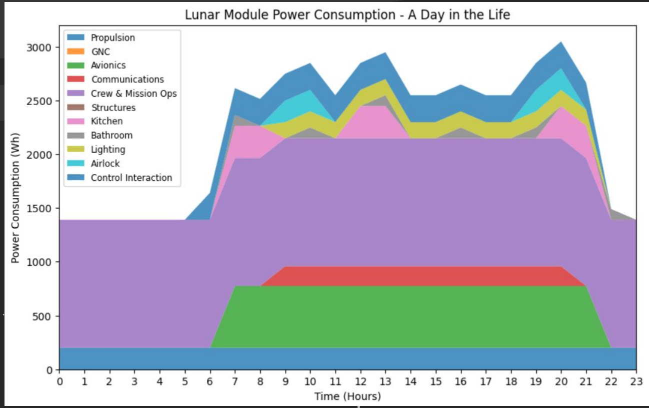
Battery Capacity Data





Graphs of Power Consumption Vs Generated

Power



Graph Of Day in Life Power Consumption

Day in Life Schedule for Reference

Total Power Consumed in 24 hour Surface Operations
53 kWh

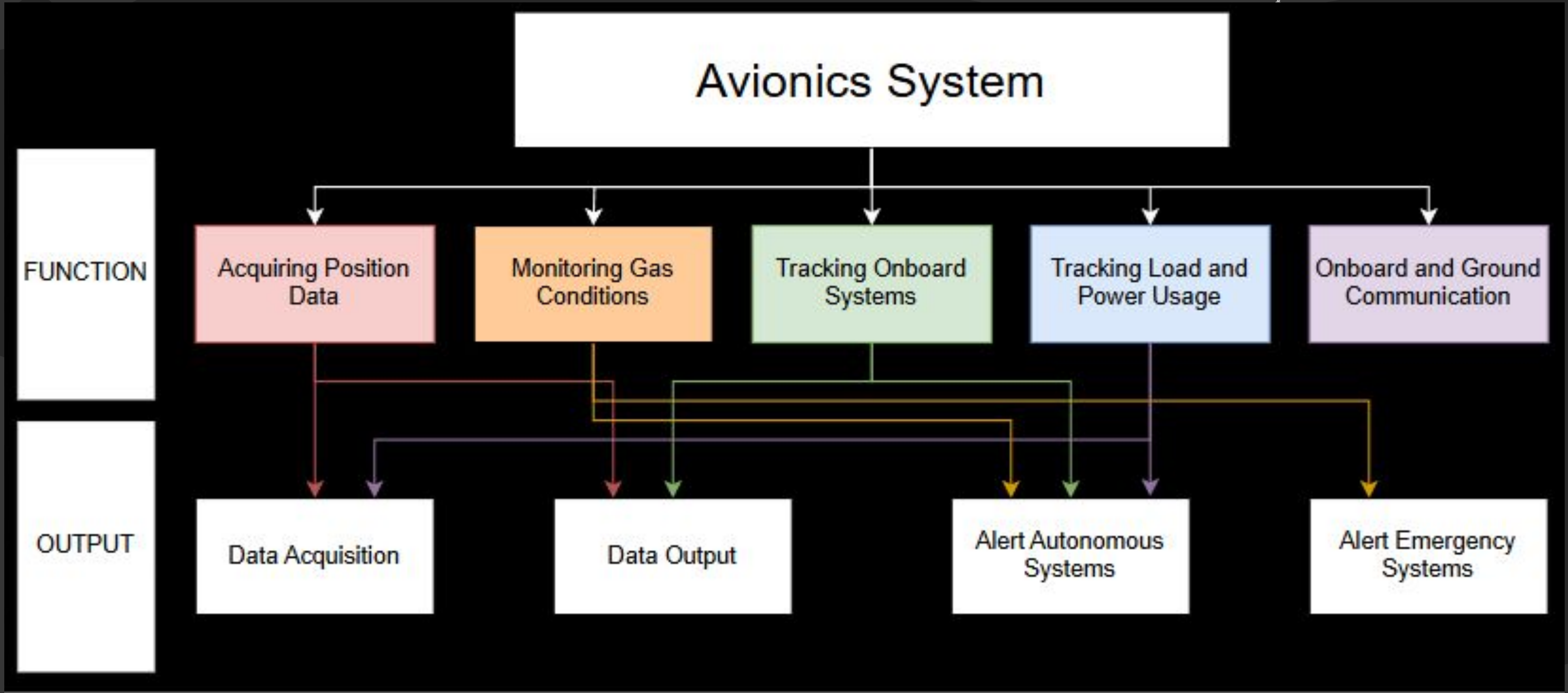
VALOR Power Draw in 24 hours

Avionics

Avionics

Traceability	Requirement	V&V
R.4	Valor shall have 4 computers for redundancy	Analysis
R.2	Valor shall include sensors to ensure optimal living conditions	Analysis
R.7	Valor shall include hardware and software for modern navigation	Analysis

Avionics


Avionics System
FUNCTION
Acquiring Position Data
Monitoring Gas Conditions
Tracking Onboard Systems
Tracking Load and Power Usage
Onboard and Ground Communication
OUTPUT
Data Acquisition
Data Output
Alert Autonomous Systems
Alert Emergency Systems

Avionics

Computing System:

- Increased Redundancy (4 Computers total, 3 main computers following Apollo Model, and 1 redundant computer)
- Processor heavy computers for less power consumption and adequate computing power
- Complete list of computer design parts found in CDR Report

Computers Trade Study

Options	Memory (Gb)	Processing Speed (GHz)	Storage (TB)	Estimated Power Draw (Watts)	Estimated Mass (kg)
Original Computers	2048*10^-6	4.3*10^-5	3.2*10^-8	55	31.75
Processor-Heavy	64	3.5	5	200	6.80
Storage-Heavy	16	1.5	10	320	9.98
Balanced	32	2.5	7	260	8.16

Avionics

Subsystem	Sensor & Display Needs
Crew Systems	O2 & CO2 saturation, O2 remaining, H2O & H2 levels, temperature, pressure
Power	Efficiency, battery recharge cycle, voltage, current, humidity, pressure, temperature, flow rate, gas leak, dust accumulation
GN&C	Gyroscope, predicted location, topographic map generator, star tracker
Propulsion	Load cells, temperature, gas pressure
Avionics	Processor, graphics card, RAM, storage, assorted parts

Avionics

Electrochemical Sensor
Crew Systems

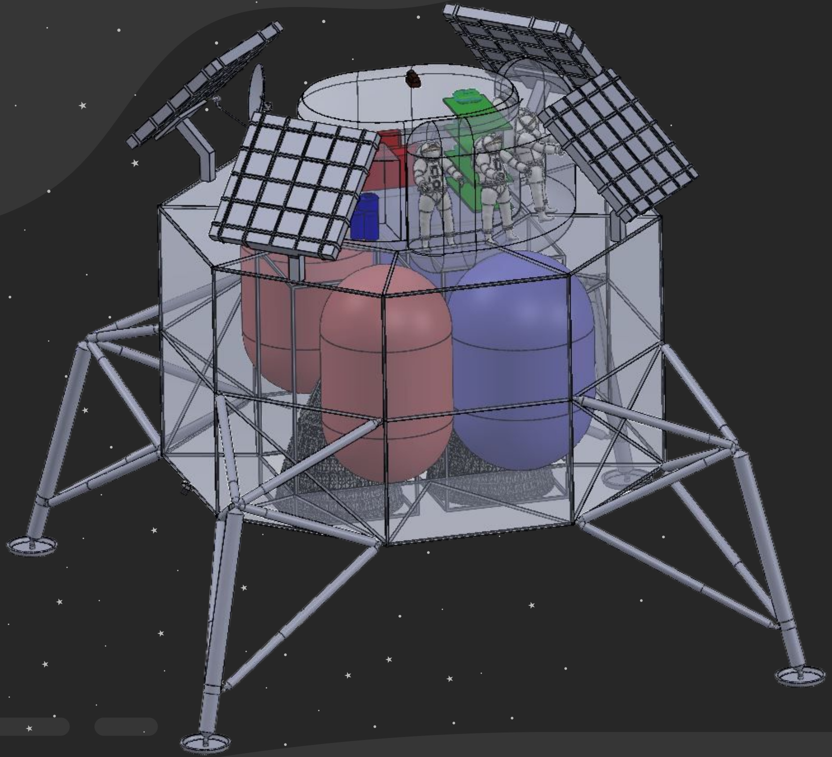
General Status Sensor
Power

Desired Measurement	Sensor
Crew Systems	
O2 Saturation	Gravity: Electrochemical Oxygen Sensor (0-100% Vol)
CO2 Saturation	Amphenol SGX Sensortech IR11EJ
Propulsion	
Gas Pressure	T200 Pressure Transducers with Cable Connection
Power	
Power Efficiency, Battery Recharge Rate, Battery Temperature	PowerMon-W - WiFi / IoT Advanced Battery Monitor / DC Power Meter with data logging
Battery Humidity	RHT-P10 Humidity and Temperature Transmitter with Remote Sensor
Water Production	Digiten G3/4"Male Thread Water Flow Hall Sensor Switch Flowmeter Counter 1-30L/min

Infrared Sensor
Crew Systems

Specialized Humidity Sensor
Power

Phase 2 Summary



Total Dry Mass:
9,441 kg

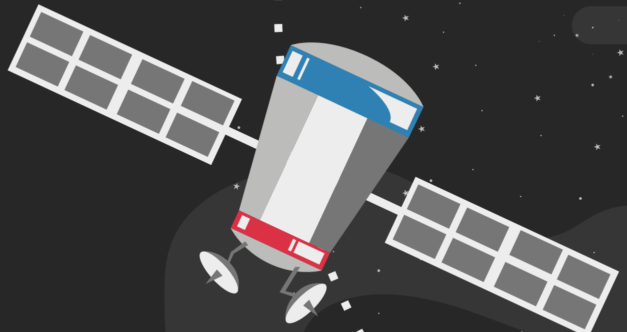
Total Wet Mass:
36,117 kg

Total Power:
723,000 Wh

Presenter: Camden Hill

04

Prototyping



Risk Assessment

Likelihood	Risk Matrix				
5			2		
4			3, 5		
3				4	1
2					
1					
Consequence	1	2	3	4	5

Risk ID	Summary
1	Structural Failure/Landing
2	Human Factors
3	Lunar Dust
4	Radiation
5	Budget Overruns

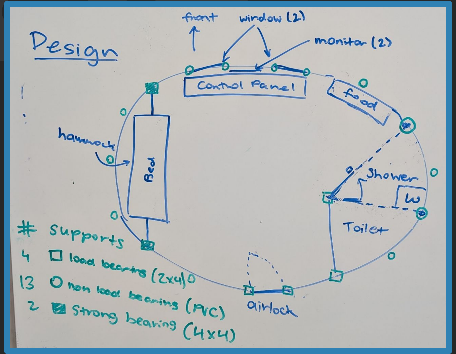
Risk Assessment

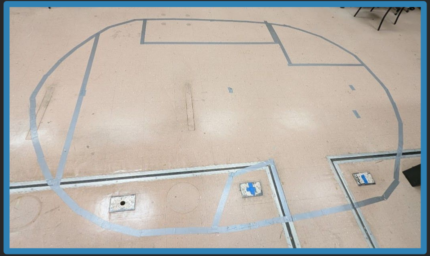
Likelihood	Risk Matrix				
5					
4		3	5		
3		4	2		
2					1
1					
Consequence	1	2	3	4	5

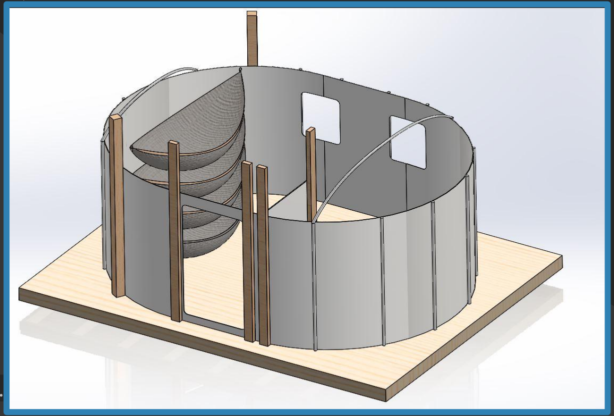
Risk ID	Mitigation Method
1	Testing of Landing Gear
2	HITL Testing
3	Airlock Procedures & Coating
4	HDPE Shielding
5	Scheduling

Presenter: Ethan Auringer

Informing Crew Cabin Design







- Address human factors risk through HITL studies
- Build to-scale crew cabin mockup
- Support key functions:
  - Sleeping
  - Eating
  - Controls
  - Storage
  - Waste Management

Build Process

Supports & Flooring




Walls & Roof

Interior Elements

Presenter: Ethan Auringer

Crew Cabin Build

Built to conduct human-in-the-loop testing & evaluation studies

AEROSPACE ENGINEERING

Testing

HITL Testing:

- 24-hour "Lunar Stay"
- 4 Crew Members
- Scheduled Group and Individual Tasks
- Contact with "Ground"
- Simulated Emergencies

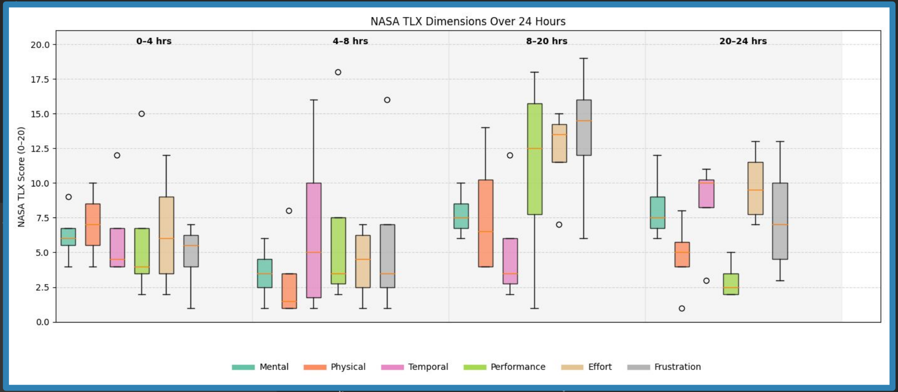
Evaluation Studies:

- 20-30 min Simulation Time
- 3-4 Volunteer Crew Members
- 4 Tasks that Allow Crew to Get a Feel for the Space
- Filled Out User Input Surveys
- Completed NASA Task Load Index & System Usability Scale

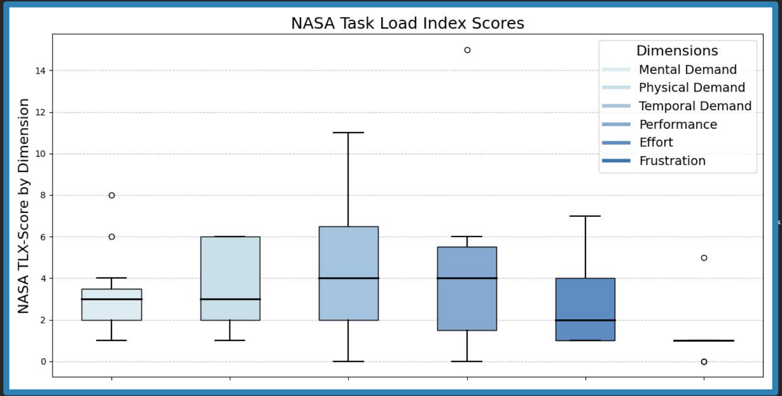
Presenter: Abby Rajagopal

Results: NASA Task Load Index

HITL Testing



Evaluation Studies



Design Changes to Crew Cabin

- Reduce the Size of the Hatch
  - Install Better Handles
- Add Two More Chairs for Crew Down Time
- Increase Lighting and Add Specific Task Lights
- Use Better Insulating Hammocks or Transition to Cots
- Possible Private Work/Conference Room
- Designate Space and Add Equipment for Exercise

Future Work: Increasing Fidelity

Given more time and resources, what would we do to increase fidelity?

- Longer Testing (3-7 Days, eventually full 2 week mission)
- Install water, plumbing, and lighting systems to reduce need to leave crew cabin.
- Isolate crew cabin mockup from outside world
- Construct an airlock and perform Don/Doff procedures
- Higher fidelity emergency simulations
- Adjustable walls to allow for quick reconfiguration and more frequent tests
- Better Materials

Presenter: Lucas Schuh

05

Cost & Schedule

Mission Cost Assessment

Surface Level Cost Analysis

Phase	Cost (USD)
Design, Development, Test & Evaluation	$2.61 Billion

Assumptions

- 2 year project
- Using already existing facility to refit the LM/VALOR
- Assembly, labor costs, launch and mission are only recurring costs

Deep Level Cost Analysis

Phase	Cost (USD)
Research & Development	$979 Million
Procurement & Delivery	$910 Million
Assembly & Testing	$920.6 Million
Satellite Constellation	$60 Million
Total (Estimate)	2.87 Billion

*does not include cost of launch and mission ops

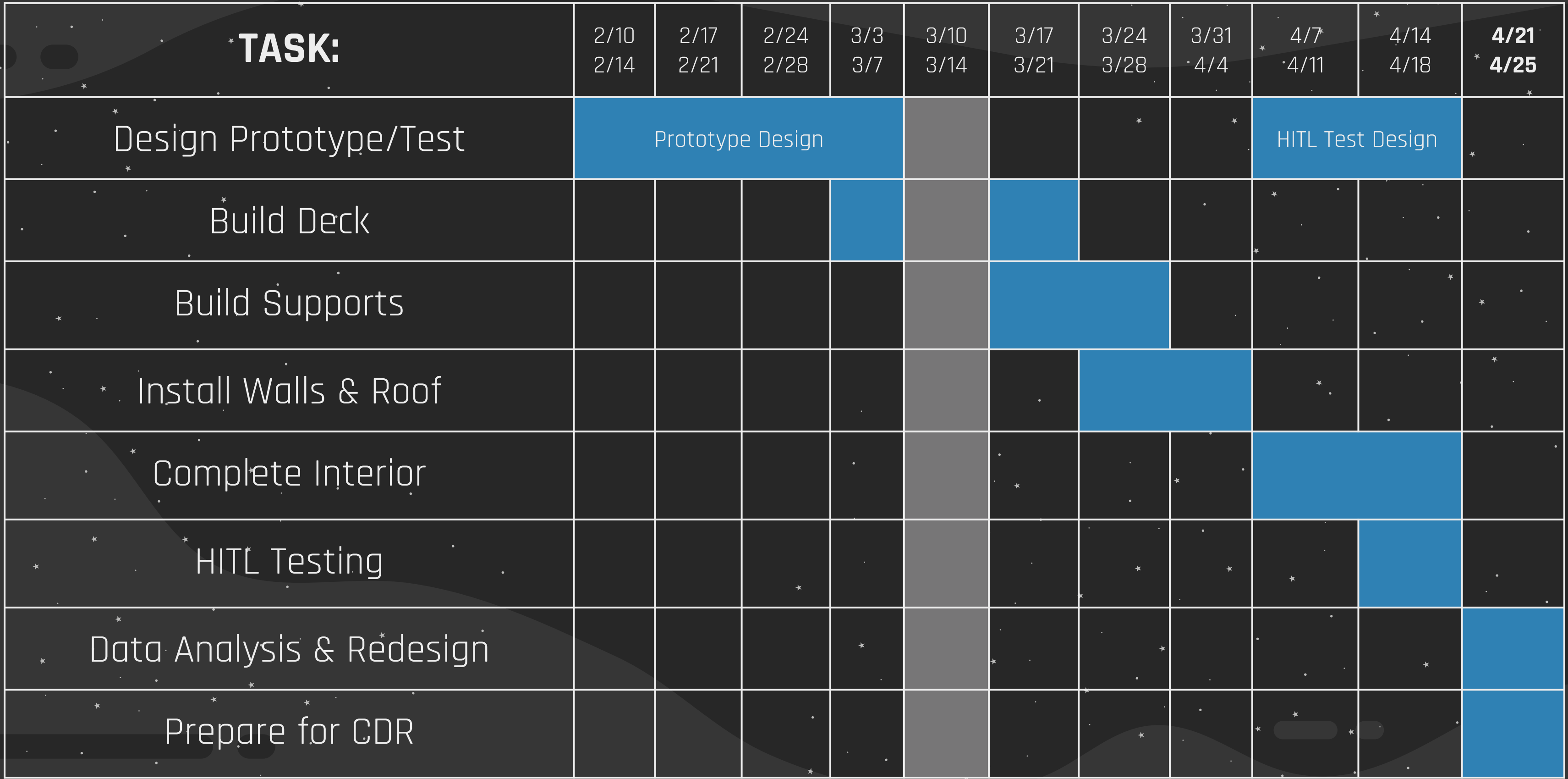
Prototype Cost Assessment

Prototype Cost Analysis	
Category	**Cost (USD)**
Deck	$324.69
Support & Walls	$598.75
Interior	$489.69
Testing	$142.87
Misc.	$45.98
TOTAL	**$1,651.98**

Design Gantt Chart

TASK:	1/13 1/17	1/20 1/24	1/27 1/31	2/3 2/7	2/10 2/14	2/17 2/21	2/24 2/28	3/3 3/7	3/10 3/14	3/17 3/21	3/24 3/28	3/31 4/4	4/7 4/11	4/14 4/18	**4/21 4/25**
Communication	Verify Design														CAD
GN&C	CAD & Design Changes														Simulations
Structures	Design Frame & Lander Legs												Leg Adjustability & Stress Analysis		
Propulsion	Tank Sizing														
Crew Systems	Verify Resource Generation									Airlock Design					
Power	Power Source Trade Study						Verify Power Requirements								CAD
Avionics	Familiarize with & Verify Design														

Prototype Gantt Chart



Acknowledgements

Academia

- Prof. John Connolly
- Devin Jain

Industry

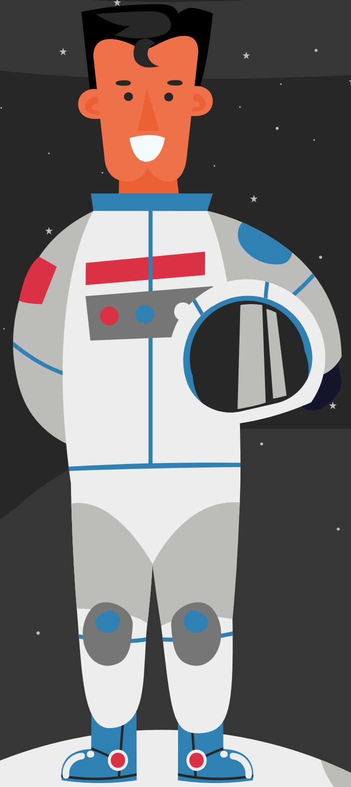
- Dr. Robert Howard (NASA)
- David Gabriel (David Gabriel Design LLC)
- Christopher Gerty (Leidos)

Shoutouts

- HRBB Facilities
- DBF Capstones
- Evaluators
- Home Depot
- Amazon Prime

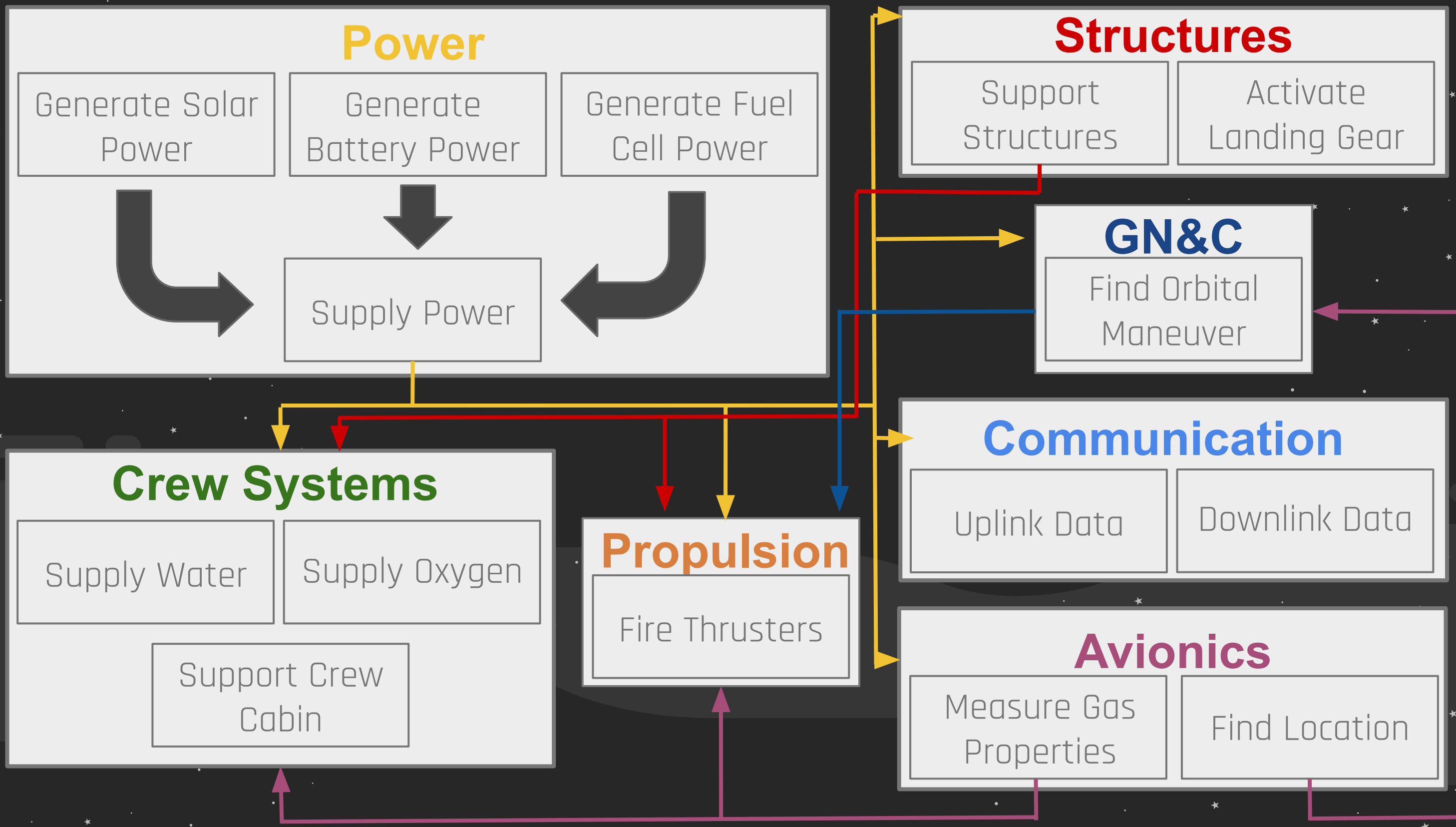
Thank you!

Questions?



BACKUP SLIDES

Subsystem Interactions

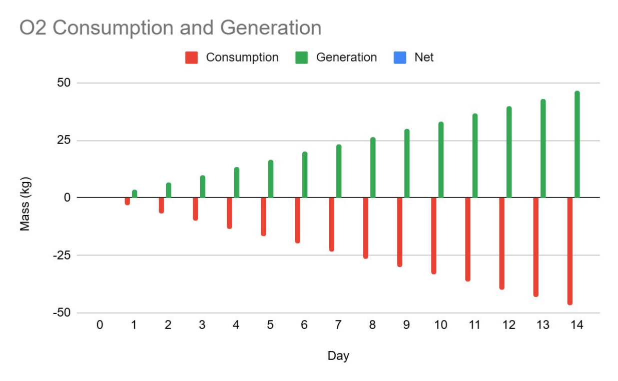


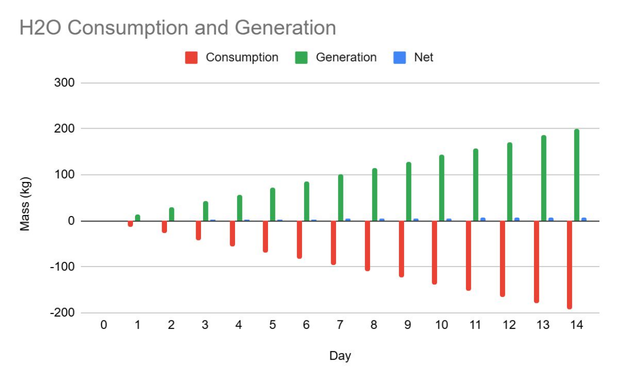
Crew Systems Trade Studies

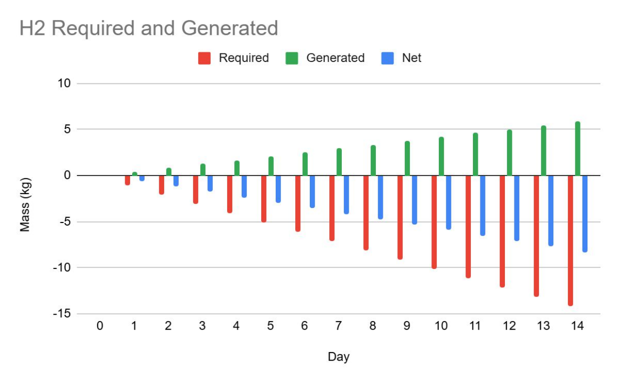
Coolant Trade Study							
Options	**Water Ratio**	**Toxic**	**Thermal Conductivity (W/mk)**	**Heat Capacity (cal/gmK)**	**Freezing Point (°C)**	**Boiling Point (°C)**	**Viscosity @ 20°C (mPa s)**
Water	N/A	no	0.61	1	0	100	1.002
Propylene Glycol	1:2	no	0.38	0.73	-59	188	2.94
Ethylene Glycol	1:2	yes	0.41	0.71	-13	197	2.13
Methanol	1:2	yes	0.49	0.8	-97	64	1.48

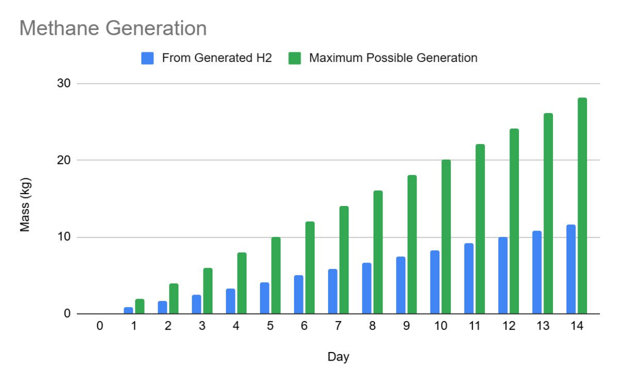
CO2 Scrubbing Trade Study					
Reactants	**Reaction**	**Reactant Mass (kg/day/crew)**	**Reactant Volume (cm^3)**	**H2O Produced (kg/crew/day)**	**O2 Produced (kg/crew/day)**
Lithium Hydroxide	$2LiOH(s) + CO_2(g) \rightarrow Li_2CO_3(s) + H_2O(g)$	1.5	1027.397	0.5642	N/A
Lithium Peroxide	$Li_2O_2(s) + CO_2(g) \rightarrow Li_2CO_3(s) + 1/2\ O_2(g)$	1.437	621.980	N/A	0.5010
Sodium Hydroxide	$2NaOH(s) + CO_2(g) \rightarrow Na_2CO_3(s) + H_2O(g)$	2.505	1176.071	0.5642	N/A

Crew Systems Trade Study









Criteria	Silver-Zinc (LM's Battery)	Amprius Silicon Nanowire Li-Ion	Lithium-Ion	Nickel-Hydrogen	Nickel-Metal Hydride	Silver-Cadmium
Energy Density (Wh/Kg)	100-120	500	100-160	60-70	60-70	60-70
Voltage per Cell	1.50	3.45	2.5-3.4	1.30	1.30	1.10
Charge/Discharge Efficiency (%)	70-75	95+	90-95	80-90	80-90	65-70
Charging Time (Hrs)	N/A	<1	2-3	8-10	4-6	4-6
Recharge Cycles	1	10000	2000-4000	10000 - 20000	5000 - 10000	1000 - 1500
Durability (Shock Resistance)	Moderate	Very High	Moderate	High	High	Moderate
Toxicity	High	Low	Low	Moderate	Moderate	High
Manufacturability	Excellent	Low	Low	Good	Moderate	Good
Shelf Life (Years)	5	10-15	10-15	12-15	10-15	7-10
Cost	High	High	Moderate	High	Moderate	High

Criteria	Thin-Film Solar Panels	Proton Exchange Membrane Fuel Cells (PEMFCs)	Vertical Solar Array Technology (VSAT)	Nuclear Fission Reactors
Energy Density (Wh/Kg)	100 - 200	800 - 1200	300-600	1000+
Power Output	1-10	1-5	5-10	10-50
Maintenance	Low	Moderate	Low	Moderate
Reliability	Moderate	Moderate	High	Very High
Lifespan (Years)	10-15	5-7	10-15	10-20
Ease of Manufacturing	High	Moderate	Moderate	Low
Costs	Low	Moderate	Moderate	Very High

Structures

	Price per Pound (USD)	Ductility	Density (g/cm^3)	Yield Strength (comp/tense) (MPa)	Stiffness (Young's modulus) (GPa)
Al 7075-T6	~1.2	11%	2.82	468/537	71
Al 2024-T4	~0.8	20%	2.77	324/468	73
Al 6069-T6	~0.5	17%	2.70	393/448	68
Al 2219-T6	~0.8	10%	2.86	289/413	73
Steel 4340	~2	22%	7.86	861/1275	210
Steel 300M	~3	11%	7.86	1585/1930	204
Stainless Steel	~0.6	60%	7.47	310/503	199
Ti-6Al-4V	~20	14%	4.43	882/951	113
Ti-10V-2Fe-3Al	~20	6%	4.65	1199/1241	110
CFRP	~7	11%	1.52	1882/3378	73
Graphite-epoxy	~11	10%	1.490	1696/2895	137

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Structures

	Density (g/cm^3)	Melting Point (C)	Emissivity	Thermal Conductivity (W/mk)
HDPE	0.930	180	93%	0.4
Lead	11.35	327.5	65%	35.3
C/C	1.38	2000	85%	233
Ceramic	2.6	3000	60%	20
PEEK	1.32	700	95%	0.29
MLI	1.2	–	90%	5
IMLI	1.2	–	90%	3

AEROSPACE ENGINEERING

NORTHROP GRUMMAN

Avionics

Computers Trade Study					
Options	Memory (Gb)	Processing Speed (GHz)	Storage (TB)	Estimated Power Draw (Watts)	Estimated Mass (kg)
Original Computers	2048*10^-6	4.3*10^-5	3.2*10^-8	55	31.75
Processor-Heavy	64	3.5	5	200	6.80
Storage-Heavy	16	1.5	10	320	9.98
Balanced	32	2.5	7	260	8.16

Propulsion

Trade Study: Raptor Engine & COPV Layout and Configuration					
Layout & Configuration	Tank Mass (kg)	Volume Efficiency	Cryogenic Layout	Radiation Protection Mass	Propellant Mass (kg)
2 Oxygen Spheres & 1 Methane Toroidal	980	Moderate	Moderate	Moderate	59,283
2 Oxygen Spheres & 2 Methane Cylinders	864	Moderate	Good	Good	58,957
2 Oxygen Cylinders & 2 Methane Cylinders (Baseline)	808	Good	Good	Very Good	58,795
2 Oxygen Spheres & 2 Methane Spheres	2nd Lowest	Poor	Good	Poor	2nd Lowest
1 Oxygen Sphere & 1 Methane Sphere	Lowest	Extremely Poor	Good	Extremely Poor	Lowest

Worse				Better

Propulsion

Phase 1 Trade Study: Commercially Available Methalox Rocket Engines								
Engines	Thrust	Mass	Specific Impulse	Throttle	Gimbaling Range	Modifications	Development Time	Development Cost
Custom Design	Moderate	Low/ Moderate	Moderate/ High	Moderate	Moderate	Low/ Moderate	Significant	Significant
SpaceX Raptor 3	Extremely High	High	High	Effective	High	Low	Low	Low
Blue Origin BE-4	Extremely High	Extremely High	High	Effective	Low/ Moderate	Low	Low	Low
NASA Project Morpheus	Low	Low	Low	Low	Unknown	High	Moderate	Moderate
Intuitive Machines VR900	Extremely Low	Low	Moderate/ High	Effective	Unknown	Moderate	Low	Low
Multiple IM-VR900	Moderate	Moderate	Moderate/ High	Effective	Moderate	Moderate	Low	Low

Worse				Better

Subteam	ITEM	MAX POWER	AVG POWER	Quantity	Total Power
Prop	Cryogenics	200	200		200 W
GNC	RW25	2.25	1.5	6	139.26 W
	CubeStar	0.396	0.264	2	
	STIM 320	3.75	2.5	2	
	Poseidon-4	135	90	1	
	NGAVGS	45	30	1	
	DCAM	22.5	15	4	
Avionics	Sensors				846.35W
	Oxygen remaining	0.115	0.115	2	
	Oxygen remaining	0.15	0.15	1	
	Carbon Dioxide Saturation	0.18	0.18	5	
	Water levels	0.15	0.15	2	
	Temperature	0.24	0.24	5	
	Hydrogen levels	0.115	0.115	2	
	Pressure	0.24	0.24	5	
	Power Efficiency/Battery Recharge Cycle/Battery Humidity	0.1272	0.1272	1	
	Bus Voltage and Current	0.2	0.2	7	
	Battery Humidity	0.6	0.6	1	
	Irradiance	0.6	0.6	4	
	Pressure	0.115	0.115	2	
	Flow Rate	0.25	0.25	2	
	Water Production	0.25	0.25	1	
	Fuel Cell Temperature	0.24	0.24	5	
	Gas Leak	0	0	5	
	Temperature (Propulsion)	0.6	0.6	1	
	Gas Pressure (Propulsion)	0.115	0.115	2	
	Processor	250	200	3	
	Graphics Card	90	70	3	
	RAM	8	4.6	3	
	Storage	3.6	3.6	3	
Comms					492 W
	Camera	7	7	2	
	Transceiver	70	70	2	
	Inflight Antenna	315	315	2	
	Erectable Antenna	100	100	1	
Crew					1189.5 W
	Coolant Pump	275	275	1	
	RCRS	530	530	1	
	Toilet	343	255.5	1	
	Water Recovery System	50	12.5	1	
	Compressors	100	100	2	
	Interior Lighting	13	7.5	6	
	Exterior Lighting	18	9	8	
Structures					10
	Actuation motors	10	0.07	4	

AEROSPACE ENGINEERING

NORTHROP GRUMMAN

Upgraded LM	
Component	Mass (Kg)
Propulsion	
Fuel	18399.44735
Liquid Oxygen Tanks	223.9416305
Liquid Methane Tanks	199.5122314
Helium Tank	22.64
Liquid Helium	20
SpaceX Raptor 3's	3050
Tubing, Valving, Cooling, & RCS (TBR)	120

Communications	
Advantech Wireless AWMT-3000K	32
BAE Systems Ku-Band SATCOM Antenna	84
HAWKEYE 4 Lite Antenna	80
Photosonics Remote Head Camera	1.1

Battery (Amprius Silicon Nanowire Li-Ion)	300
Thin Film Gallium Arsenide Solar Panels	40
Redwire's Roll-Out Solar Arrays (ROSA)	49

Structures	
Body Group	2785.59
Landing Gear	1008.3
Thermal Shielding	87.3

Crew Systems	
HAM STD 190 PKG	46.18
190 PKG HDW	4.54
O2+H2O COOLANT ASSY	16.37
ATMOS REVIT SECTION	1.27
LI2O2 CARTRIDGES	80.47

Cabin Atmosphere Supply and Cabin Pressure Control	
PRESSURIZATION SYSTEM	12.07
PRESSURIZATION SYSTEM	2.04
PLUMBING-PRESS SYS A	5.4
PLUMBING-PRESS SYS B	5.31
TOTAL TANKS	38.64
390 OX MODULE	4.04
390 PKG-HDW	3.49
PRESSURIZATION PLUMBING	30.89
TANKED OXYGEN	64
TANKED NITROGEN	14.84
PLSS O2 RECHARGE	1.59

GN&C	
iADCS-200	0.84
RW25	1.2
CubeStar	0.11
STIM 320	0.114
Poseidon-4	60
NGAVGS	8
DCAM	9.6

Water Management Section	
490 H2O MODULE	2.59
490 PKG-HDW	2.22
TANKED WATER	206.8
H2O SYSTEM	10.25
UWMS	90

Heat Transfer Section	
-290 SUBTOTAL	3.58
HTS PRI LOOP	33.88
HTS SECONDARY LOOP	10.89
COLD PLATES R&D	2.63
TOTAL PRIMARY CP	1.18
GLYCOL SYSTEM	2.72

Other	
OUTER LIGHTING	20
INNER LIGHTING	5
WASTE MANAGEMENT	1.77
PLSS BATTERIES	10.16
BPS INSTALLED HARDWARE	1.04
CREW PROVISIONS	99.97
FURNISHINGS	60
EMUs	200

Avionics	
Intel - Core i7-14700K	0.3
NVIDIA RTX A2000 12GB 4xmDP Graphics Card	6
Lenovo 64GB DDR5 4800MHz ECC RDIMM Memory	0.12
SAMSUNG 990 PRO SSD 4TB PCIe 4.0	0.054
Structural Components	15
Total Computer Mass	21.474
Gravity: Electrochemical Oxygen Sensor (0-100% Vol)	1.792
T200 Pressure Transducers with Cable Connection	0.14
Industrial Liquid Level Sensor MODBUS RS485 Aviation Connector	0.67
Amphenol SGX Sensortech IR11EJ	0.15
Industrial Liquid Level Sensor MODBUS RS485 Aviation Connector	1.34
AH Temperature and Humidity Sensor - 4m Cable	0.33
NP785 Ultra Low Differential Pressure Transmitter	0.3
Total Sensor Mass	4.772

Propulsion	3212.64
Communications	197.1
Power	403.6
GN&C	79.864
Structures	3881.19
Crew Systems	1095.82
Avionics	52.442
Dry Mass (No Tanks)	**8922.656**
Dry Mass (W/ Tanks)	**9440.753733**
Wet Mass	**36117.14491**
36,117	

GN&C

	Hardware	Number of Units	Mass	Power
Integrated Unit	iADCS-200	2	840 g	N/A
Reaction Wheels	RW25	6	1.2 kg	4.5 W
Star Tracker	CubeStar	2	110 g	0.264 W
IMU	STIM 320	2	114 g	2.5 W
Landing Radar	Poseidon-4	1	60 kg	90 W
Docking Radar	NGAVGS	1	8 kg	30 W
Docking Camera	DCAM	2	4.8 kg	15 W
Landing Camera	DCAM	2	4.8 kg	15 W
Total	**N/A**	N/A	**79.9 kg**	**157.3 W**

GN&C

	Weight	RW3-0.06	VRW-D-2	Trillian-1
Mass	0.90	8	2	4
Momentum Capacity	0.75	3	10	7
Peak Torque	0.75	3	4	10
Power	0.50	5	2	9
Cost	0.30	5	5	5
Total	32	15.7	14.8	22.4

	RW3-0.06	VRW-D-2	Trillian-1	RW25
Mass	235 g	2 kg	1.5 kg	200 g
Momentum Capacity	0.18 Nms	2 Nms	1.2 Nms	0.03Nms
Peak Torque	0.02 Nm	0.05 Nm	47.1 Nm	0.002 Nm
Power	23.4 W	65 W	24 W	1.5 W
Cost	$80,000	~ $80,000	~ $80,000	~ $40,000

GN&C

	Weight	CubeStar	ST-1	ST200
Mass	0.90	8	5	10
Cross Axis/Twist Accuracy	0.85	9	4	8
Field of View	0.75	9	6	6
Power	0.50	9	3	7
Cost	0.20	5	5	5
Total	32	27.1	14.9	24.8

	CubeStar	ST-1	ST200
Mass	55 g	108 g	40 g
Cross Axis/Twist Accuracy	55.4 in, 77.4 in	8 in, 50 in	30 in, 200 in
Field of View	58° x 47°	21° x 21°	22° x 22°
Power	0.264 W	1.2 W	0.65 W
Cost	~ $ 60,000	~ $ 60,000	~ $60,000 (Quote)

GN&C

	Weight	VN-100	VN-110	STIM 320
Mass	0.90	9	3	6
Gyro/Accelerometer Bias	0.85	3	7	10
Degrees of Freedom	0.75	6	6	6
Power	0.50	9	6	6
Cost	0.20	8	6	6
Total	32	21.3	18.3	22.6

	VN-100	VN-110	STIM 320
Mass	15 g	125 g	57 g
Gyro/Accelerometer Bias	10°/ hr, 40 µg	1°/hr, 10 µg	0.3°/ hr, 3 µg
Degrees of Freedom	3	3	3
Power	0.22 W	2.5 W	2.5 W
Cost	$ 500	~ $1,000 (Quote)	~ $1,000 (Quote)

GN&C

	Weight	NASA NRA	ATLAS	Poseidon-3C	Poseidon-4
Mass	0.90	5	2	9	10
Power	0.80	5	2	10	9
Range	0.75	8	3	5	8
Cost	0.60	3	1	8	9
Accuracy	0.40	4	10	2	6
Antenna Length	0.35	8	3	6	9
Total	38.5	20.7	11.3	27.6	33.2

	NASA NRA	ATLAS	Poseidon-3C	Poseidon-4
Mass	230 kg	470 kg	70 kg	60 kg
Power	237 W	420 W	78 W	90 W
Range	1,336 km	496 km	890 km	1,336 km
Cost	$175 M	$760 M	$5 M	$ 4 M
Accuracy	2.4 cm	0.2 cm	3.4 cm	1.2 cm
Antenna Length	1.5 m	17 m	5 m	1.2 m

GN&C

	iADCS-200
Mass	470 g
Actuators	3 reaction wheels, 3 magnetic torques
Size	95 x 90 x 32 mm
Sensors	1-star tracker, 1 IMU, 3 RW
Pointing Accuracy	<1°
Cost	$ 45,000

	DCAM
Mass	2.4 kg
Power	12 W
Range	5 km
Accuracy	2592 px, 350 nm - 850 nm wavelength
Size	13.6 cm x 23.1 cm x 18.3 cm

	NGAVGS
Mass	8 kg
Power	30 W
Range	5 km
Accuracy	Uses two light sources and their wavelengths to determine position and velocity
Size	17.8 cm x 19 cm x 30.5 cm

Communications

	S-Band Transceiver	Anasat Ku-Band 2W	Emcore Optiva OTS-2	Advantech Wireless AWMT-3000K
Bit Rate	51.2 kBPS	16 MBPS	500 MBPS	800 MBPS
Range Rating	3,000 km	35,000 km	35,000 km	35,000 km
Mass	65 kg	26 kg	40 kg	22 kg
Power	70 W/hr	60 W/hr	120	100

Table 4.1.2.1 *Transceiver Trade Study*

	S-Band Steerable Antenna	KuStream 1000 Antenna System	Ku Band SATCOM antenna
Mass	13 kg	44 kg	22 kg
Power	40 W/hr	115 W/hr	315 W/hr

Table 4.1.2.3 *Inflight Antenna Trade Study*

	S-Band Erectable Antenna	HAWKEYE 4 Lite Antenna	CPI SAT 1259 2.4M Flyaway
Mass	6 kg	38 kg	65.7 kg
Power	20 W/hr	90 W/hr	100 W/hr

Table 4.1.2.1 *Erectable Antenna Trade Study*

	Portable TV Camera	Space Eye SE 320	Photosonics Remote Head Camera
FPS	10	180	30
Mass	3.3 kg	2 kg	1.1 kg
Power	6.25 W/hr	8.7 W/hr	15 W/hr

Table 4.1.2.4 *Camera Trade Study*

CSC Cost Estimate

Cost of 2000 GEO Satellite: 20000 mil
https://www.researchgate.net/figure/Comparison-of-cost-of-satellite-constellations-with-10-million-small-satellites_tbl5_347276446

1 Satellite = 10 mil

TOTAL: 60 mil